# Post-Earnings-Announcement Drift: Mechanism, Evidence, and a Hedged Equity–Options Strategy

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**Abstract**

Markets under-react to earnings news: a stock keeps moving in the direction of its earnings surprise for weeks after the announcement. We model this as gradual incorporation of information by heterogeneously attentive investors and test the model's predictions in a 182,712-event sample (1996–2013): the top-minus-bottom surprise-decile spread is +3.05 % over 60 trading days ($t$ =5.25, clustered by announcement quarter) and +3.76 % after a size and book-to-market adjustment ($t$ =9.25); the same ordering appears in option premia, amplified about 4.7 ×. We then ask how much can be traded, and we audit the answer. The *validated* market-neutral equity book `calm_weighted` (positions sized by inverse volatility) earns 6.71% a year at 4.49% volatility, a Sharpe ratio of 1.47 and a maximum drawdown of −15.6%; the *research* book `calm_rank_tilt`, whose tilt exponent was chosen after seeing the results, earns 7.12% at a Sharpe ratio of 1.81 (drawdown −11.6%). Against the CRSP value-weighted market, `calm_weighted` has an annualized alpha of 6.9% ($t$ =6.1) and a beta of 0.00. The corrected Kelly options books earn 4.12% (`options_cap20`, Sharpe 1.77) and 5.69% (`options_cap30`, Sharpe 1.55); an earlier convention for marking expired quotes had overstated the first at 8.58% and a Sharpe ratio of 3.78. The two sleeves are nearly uncorrelated (daily correlation −0.05); the validated blend `hedged_blend` earns 11.33% at 4.83% volatility (Sharpe 2.25, drawdown −16.7%) and its research counterpart `calm_blend` earns 12.43% (Sharpe 2.50). The edge is concentrated in one kind of trade (a long position in a thinly traded, low-beta stock held for more than two months), high-beta shorts earn no measurable alpha, and alpha per dollar has fallen by a factor of 3.9 between 1996–2001 and 2008–13. Several corrections are identified but not yet applied to these numbers (Section 10), and borrow fees, financing, market impact and option spreads beyond a one-tick floor are not modelled (Section 11). All figures are hypothetical backtests.

**Summary of results (backtest, 1996–2013, $10M initial capital, net of modeled costs)**

| Book | Return | Vol | Sharpe | Max DD | Status |
|---|---|---|---|---|---|
| `calm_weighted` (equity) | 6.71% | 4.49% | 1.47 | −15.6% | validated |
| `calm_rank_tilt` (equity) | 7.12% | 3.83% | 1.81 | −11.6% | research |
| `balanced_leverage` (equity) | 6.49% | 4.94% | 1.30 | −16.6% | reference |
| `options_cap20` | 4.12% | 2.30% | 1.77 | −13.8% | validated |
| `options_cap30` | 5.69% | 3.61% | 1.55 | −22.3% | validated |
| `hedged_blend` | 11.33% | 4.83% | 2.25 | −16.7% | validated |
| `calm_blend` | 12.43% | 4.73% | 2.50 | −15.5% | research |

Statistics are computed over each book's full calendar, as stored in `results/leaderboard.md`. Leveraged and market-exposed research variants are in Table 7. All figures are hypothetical backtest results.

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# 1 Introduction

When a firm reports earnings that differ from expectations, its price does not adjust fully on the announcement. Over the following weeks it keeps moving in the direction of the surprise (Ball and Brown, 1968; Bernard and Thomas, 1989, 1990). This post-earnings-announcement drift (PEAD) is attributed to underreaction and investor inattention (Hong and Stein, 1999; Hirshleifer et al., 2009; DellaVigna and Pollet, 2009), and it survives because the capital that could remove it is costly to deploy (Shleifer and Vishny, 1997; Mendenhall, 2004).

This paper has one thesis: PEAD is a gradual transfer of value from slow to fast investors, and it can be harvested in two structurally different ways. It also has a second purpose, which is an audit. The strategies were built, rebuilt and renamed over several months, and earlier drafts reported numbers that later corrections lowered. The version here reports the corrected numbers, labels every in-sample choice, and lists every defect found (Section 10). The paper is organized around four questions.

(i) **Where does the money come from?** Section 2 gives a model of gradual incorporation. It predicts the shape of the drift in time, its dependence on frictions, and its amplification in options.

(ii) **Does the phenomenon exist in these data?** Sections 3 and 4 describe the sample and test every prediction of the model before any strategy is built: clustering-robust significance, size and value controls, cross-sectional heterogeneity, rival hypotheses, and confirmation in a second market.

(iii) **How well does it trade, and where does the return actually come from?** Sections 5–8 build a linear equity family and a convex options book, decompose their returns by leg, by kind of trade and by era, and evaluate them separately and together.

(iv) **How much should a reader believe the numbers?** Sections 9–11 report the capacity and risk-budget frontiers, a multiple-testing adjustment based on the trials we can count, the audit log, and what the backtest does not model.

**Names and status.** Every strategy has a plain name, and a one-line description at its first use. Books are labelled *validated* (parameters fixed before the result, no look-ahead we know of), *research* (some parameter or design choice was made after seeing results, or the book is leveraged or market-exposed), or *reference* (kept as a comparison, including the overstated legacy options books). Earlier drafts and the project history used numbered labels ("Strategy 6", "opt_kelly" and so on); Table 21 in Appendix A maps them to the names used here. The headline equity result is the validated book `calm_weighted`; the research book `calm_rank_tilt` is reported beside it and flagged each time; `balanced_leverage` is the reference every volatility-sized book is compared with.

# 2 Where the money comes from

## 2.1 Gradual incorporation of a surprise

Let firm $i$ announce earnings at day 0 with standardized surprise $z_i$, normalized to have mean zero and unit variance within its announcement quarter. Suppose the surprise changes the firm's full-information value by $\lambda z_i$ (in return units, with $\lambda > 0$ the information content of a unit surprise). The price embeds only a fraction $\phi(d)$ of that revision $d$ days after the announcement, with $0 \leq \phi(0) < 1$

and $\phi(d) \uparrow 1$. Then the market-adjusted cumulative return to horizon $h$, measured from the day after the announcement, is

$$\mathbb{E}\big[\mathrm{CAR}_i(h) \mid z_i\big] = \lambda\,\big[\phi(h) - \phi(0)\big]\,z_i, \tag{1}$$

so the cross-sectional *spread* between the top and bottom surprise groups is

$$\Delta(h) = \mathbb{E}[\mathrm{CAR} \mid \mathrm{top}] - \mathbb{E}[\mathrm{CAR} \mid \mathrm{bottom}] = \lambda\,\big[\phi(h) - \phi(0)\big]\,(\bar{z}_{\mathrm{top}} - \bar{z}_{\mathrm{bottom}}). \tag{2}$$

The spread is the economic object that both strategies harvest. It is non-negative, increasing, and bounded by $\lambda(1-\phi(0))(\bar{z}_{\mathrm{top}} - \bar{z}_{\mathrm{bottom}})$.

**Microfoundation: heterogeneous attention.** Suppose a fraction of capital updates at an idiosyncratic Poisson rate $\kappa$, with $\kappa$ distributed across investors according to a distribution $G$. Then $\phi(d) = \phi(0) + (1-\phi(0)) \int (1 - e^{-\kappa d})\, dG(\kappa)$. If $G$ is Gamma with shape $k$ and rate $\theta$,

$$\phi(d) = \phi(0) + \big(1 - \phi(0)\big)\Big[1 - \big(1 + d/\theta\big)^{-k}\Big]. \tag{3}$$

A mixture of speeds makes incorporation *concave but slow*: fast updaters finish within days, while inattentive investors take months. A single speed would produce an exponential approach and a spread that flattens within a few multiples of $1/\kappa$. Equation (3) instead produces a spread that keeps rising at a decreasing rate, which is the shape measured in Section 4.

## 2.2 Limits to arbitrage determine where the drift is largest

Informed capital closes the gap only until the marginal profit no longer covers the marginal cost of carrying the trade. Let $\xi_i$ be the per-day all-in cost of holding a position in firm $i$ (spread, impact, borrow, and the idiosyncratic risk the arbitrageur must bear), and let $\alpha_i(d) = \lambda\phi'(d)|z_i|$ be the daily drift. Arbitrage capital flows in while $\alpha_i(d) > \xi_i$ and stops when $\alpha_i(d) \approx \xi_i$. The drift that survives is therefore increasing in $\xi_i$: it is largest in small, illiquid, hard-to-short stocks. This is a sharp, testable cross-sectional prediction, which Section 4 examines: the spread should fall with firm size. It also carries a warning that the backtest cannot remove. The places where the drift is largest are the places where the costs the backtest omits (borrow, impact) are largest, so the model predicts that the omitted costs fall hardest on exactly the trades that earn the most (Section 11).

## 2.3 Why options carry a larger signal

An option is a leveraged claim on the underlying. For a call with price $C$ and delta $\Delta_c$ on a stock priced at $S$, a small stock return $r$ changes the option's return on premium by approximately $\Omega\, r$, where

$$\Omega = \frac{\Delta_c\, S}{C} \tag{4}$$

is the option's elasticity. At the money, $C \approx 0.4\, S\sigma\sqrt{T}$, so $\Omega \approx \Delta_c/(0.4\,\sigma\sqrt{T})$. With $\Delta_c \approx 0.5$, maturity near 4 months and annual volatility $\sigma \in [0.35, 0.45]$, this gives $\Omega \approx 4.8$–$6.2$. The drift in the option's premium is therefore $\Omega$ times the drift in the stock, less a decay and volatility cost that grows with the holding period. Two further features matter. Losses on a purchased option are capped at the premium, and the payoff is convex in the stock price, so a few large moves contribute disproportionately to growth.

Table 1: Sample construction. Counts are read from `data/ff_coverage_stats.csv`, `data/option_coverage_stats.csv`, the saved options runs and the event panel.

| | |
|---|---|
| Earnings events in the signal sample (analyst-based surprise), 1996–2013 | 182,712 |
| with book equity (Compustat `ceq`$> 0$, $\geq$ 130-day reporting lag) | 175,135 |
| with both size and book-to-market (characteristic-adjusted sample) | 175,016 |
| with a complete daily price path (equity backtests) | 173,415 |
| Events with a selected near-the-money call (option evidence) | 125,499 |
| of which priced at $+60$ trading days | 108,126 |
| Events eligible for the Kelly options book (`options_cap20`) | 81,605 |
| of which funded | 65,003 |

### 2.4 Two ways to earn the same spread

The model suggests two harvesting designs.

- **Linear.** Take long positions in positive surprises and short positions in negative surprises, and hedge everything that is not the drift. Expected profit is $\sum_i w_i \, \mathbb{E}[\mathrm{CAR}_i]$ for weights $w_i$. This is a breadth strategy: profit is roughly proportional to the number of independent bets, and risk comes from residual common factors.
- **Convex.** Buy the option that gives the most favorable growth per unit of capital at risk. Each event is a bounded-loss bet. Profit again scales with breadth, but the payoff skew is positive and the maximum loss is known in advance.

The two books earn from the same event but through different return distributions. The equity book is exposed to its short leg in a junk rally, and the options book is not. This is why they diversify each other (Section 8).

## 3 Data and sample construction

**Sources.** I/B/E/S analyst consensus and actuals; CRSP daily returns and the CRSP value-weighted market return $r_{m,t}$ (`vwretd`); Compustat annual common equity (`ceq`); OptionMetrics IvyDB daily option quotes. All four are licensed; none is redistributed with the repository. The data-build stages and their dependency graph are listed by `python -m pead prep --list`.

**Events and the surprise.** An event $i$ is a firm-quarter earnings announcement with a resolved day-0 session (the first trading day at or after the announcement). The surprise is analyst-based,

$$\mathrm{SUE}_i = \frac{A_i - F_i}{s_i}, \tag{5}$$

with $A_i$ realized earnings per share, $F_i$ the latest consensus forecast, and $s_i > 0$ a firm-specific scale. Within each announcement quarter the surprise (rounded to $10^{-8}$) is ranked with average ranks for ties into a percentile $u_i \in (0, 1]$ and assigned to deciles $D_i = \min(\lceil 10u_i \rceil, 10)$. The quarterly ranking avoids multi-year look-ahead in the breakpoints but still ranks an early-quarter announcement against announcements that come later in the same quarter; this is a look-ahead we have identified and not yet removed (Section 10). The signal sample has 182,712 events across $Q = 72$ announcement quarters; the equity backtests trade the 173,415 of them that have a complete price path in the daily panel, on 6,558 firms (Table 1).

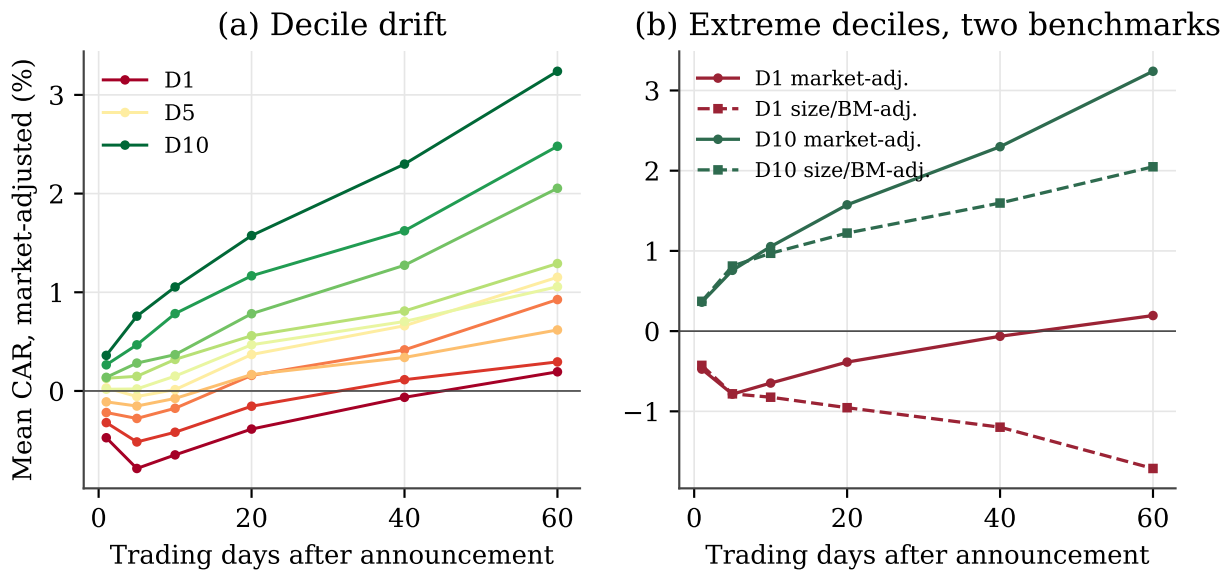



Figure 1: (a) Mean market-adjusted cumulative return by SUE decile and horizon. (b) The extreme deciles under the market benchmark (solid) and under the leave-one-out size/book-to-market benchmark (dashed). Source: `data/decile_horizon_stats.csv`, `data/ff_decile_horizon_stats.csv`.

**Returns.** For event $i$ with day-0 index $t_0$, the market-adjusted cumulative return to horizon $h$ is $\mathrm{CAR}_i(h) = \prod_{k=1}^{h}(1 + r_{i,t_0+k}) - \prod_{k=1}^{h}(1 + r_{m,t_0+k})$, with $h \in \{1, 5, 10, 20, 40, 60\}$ trading days. Accrual starts on day $t_0 + 1$, so the announcement-day reaction is not counted as drift. The equity backtests use the daily panel from 1995-01-03 to 2014-06-30; events end in 2013, and the options calendar ends in August 2013 with 2011 missing (Section 11).

# 4 Establishing the anomaly

## 4.1 Inference under earnings-season clustering

Firms announcing in the same season share common shocks, so event-level standard errors overstate precision. We use the Fama–MacBeth estimator (Fama and MacBeth, 1973) with the announcement quarter as the unit of observation. For an event-level quantity $x_i$ and event set $\mathcal{S}$, with $\mathcal{I}_q$ the events in quarter $q$,

$$\bar{x}_q = \frac{1}{|\mathcal{S} \cap \mathcal{I}_q|} \sum_{i \in \mathcal{S} \cap \mathcal{I}_q} x_i, \qquad \hat{\mu} = \frac{1}{Q} \sum_{q=1}^{Q} \bar{x}_q, \qquad \widehat{\mathrm{SE}}(\hat{\mu}) = \frac{1}{\sqrt{Q}} \Big[ \tfrac{1}{Q-1} \textstyle\sum_q (\bar{x}_q - \hat{\mu})^2 \Big]^{1/2}. \tag{6}$$

The estimator is consistent under arbitrary dependence within a quarter. The spread is formed quarter by quarter, $\Delta_q(h) = \overline{\mathrm{CAR}}_{q,D10}(h) - \overline{\mathrm{CAR}}_{q,D1}(h)$, so its standard error reflects the fact that both legs are drawn from the same cross-section.

## 4.2 The drift in equities

Table 2 and Figure 1 show the canonical pattern. Decile means are strictly increasing in surprise at +5d, +10d, +20d; at +1d, +40d, +60d the tails are ordered but the middle deciles are not strictly

Table 2: Mean market-adjusted CAR (%) by SUE decile and horizon (Fama–MacBeth means over 72 quarters).

| Decile | +1d | +5d | +10d | +20d | +40d | +60d |
|---|---|---|---|---|---|---|
| D1 (most negative) | −0.47 | −0.79 | −0.65 | −0.39 | −0.06 | +0.19 |
| D2 | −0.32 | −0.52 | −0.42 | −0.15 | +0.11 | +0.29 |
| D3 | −0.22 | −0.28 | −0.18 | +0.16 | +0.42 | +0.93 |
| D4 | −0.11 | −0.15 | −0.08 | +0.17 | +0.34 | +0.62 |
| D5 | +0.03 | −0.06 | +0.01 | +0.37 | +0.66 | +1.15 |
| D6 | +0.02 | +0.02 | +0.15 | +0.47 | +0.70 | +1.06 |
| D7 | +0.13 | +0.15 | +0.32 | +0.56 | +0.81 | +1.29 |
| D8 | +0.14 | +0.28 | +0.37 | +0.78 | +1.27 | +2.05 |
| D9 | +0.26 | +0.47 | +0.78 | +1.17 | +1.62 | +2.48 |
| D10 (most positive) | +0.36 | +0.76 | +1.05 | +1.57 | +2.30 | +3.24 |

monotone. The top-minus-bottom spread rises with every step in the horizon, from 0.83 % at one day to 3.05 % at 60 days (Table 3), and the smallest clustered $t$-statistic across all horizons and both benchmarks is 5.2. A one-time reaction that merely differed across deciles would produce a flat spread; a rising spread is the signature of gradual incorporation.

**The shape of the drift.** Fitting $\Delta(h) = a\,h^b$ to the six horizons gives $\Delta(h) \approx 0.82\,h^{0.35}$ percent for the characteristic-adjusted spread ($R^2 = 0.98$ in logs) and $0.87\,h^{0.29}$ for the market-adjusted spread ($R^2 = 0.97$). A single-speed saturating exponential, $A(1 - e^{-h/\tau})$, fits visibly worse on the same points ($R^2 = 0.78$ and 0.71 in levels, against 0.96 and 0.96 for the power law). Growth that is a power of the horizon and has not saturated by day 60 is consistent with the heterogeneous-attention mixture (3), in which the unincorporated fraction decays as a power and there is no single time scale. With only six horizons the evidence is suggestive, not a test between the two functional forms.

### 4.3 Is the drift a risk premium in disguise?

Extreme-bad-news firms tend to be small, high book-to-market firms that earn a size and value premium. In market-adjusted terms the worst decile's market-adjusted return recovers to 0.19% by day 60, which raises the possibility that part of the drift is compensation for risk rather than mispricing. We test this with a characteristic benchmark in the spirit of Daniel et al. (1997). Each event is assigned to a cell $c(i)$ defined by quarter, size quintile, and book-to-market tercile, and its return is measured against the average of its cell-mates, excluding itself:

$$\widetilde{\mathrm{CAR}}_i(h) = \mathrm{CAR}_i(h) - \frac{\sum_{j \in c(i)} \mathrm{CAR}_j(h) - \mathrm{CAR}_i(h)}{n_{c(i)} - 1}. \tag{7}$$

Leaving the event out of its own benchmark prevents its return from shrinking toward zero in thin cells. If the drift were a risk premium, the adjustment would shrink the spread. It enlarges it: the 60-day spread moves from 3.05 % to 3.76 % ($t = 9.25$), and the worst decile no longer recovers: its size/book-to-market-adjusted 60-day return is −1.71%. The apparent reversal of the worst decile was a size/value premium sitting on top of the drift (Table 3, Figure 3).

Table 3: Top-minus-bottom decile spread by horizon (%), with quarter-clustered $t$-statistics. Equity columns use the full sample. Option columns use the option-matched subsample (67 quarters; 65 at +60d) and measure the percentage return on premium of a buy-and-hold near-the-money contract at its mid price.

| | Mkt-adj. equity | | Size/BM-adj. equity | | Call | | Put | | Straddle | |
|---|---|---|---|---|---|---|---|---|---|---|
| Horizon | $\hat{\mu}$ | $t$ | $\hat{\mu}$ | $t$ | $\hat{\mu}$ | $t$ | $\hat{\mu}$ | $t$ | $\hat{\mu}$ | $t$ |
| +1d | 0.83 | 10.90 | 0.80 | 11.41 | 4.01 | 13.33 | −2.49 | −13.21 | 0.27 | 3.31 |
| +5d | 1.54 | 12.23 | 1.59 | 14.21 | 8.03 | 13.27 | −5.11 | −13.04 | 0.77 | 4.61 |
| +10d | 1.70 | 10.17 | 1.79 | 12.30 | 9.54 | 11.87 | −5.86 | −13.01 | 1.24 | 4.87 |
| +20d | 1.96 | 8.74 | 2.18 | 12.09 | 11.29 | 8.34 | −6.46 | −7.91 | 2.18 | 4.97 |
| +40d | 2.36 | 5.64 | 2.79 | 9.68 | 11.76 | 5.67 | −8.27 | −6.64 | 2.80 | 5.10 |
| +60d | 3.05 | 5.25 | 3.76 | 9.25 | 14.38 | 4.69 | −10.79 | −6.29 | 3.98 | 5.22 |

### 4.4 The cross-section of the drift

If frictions sustain the drift, it should differ across stocks in a predictable way. We compute the 60-day quarterly spread separately within each group and test equality of group means with a Cochran $Q$ statistic built from the group estimates and their standard errors. This treats groups as independent, which they are not exactly (they share quarters), so the $p$-values are approximate. The one-way ANOVA on the raw spread series reported in earlier drafts needs the series themselves, which are not among the saved files; we therefore do not quote it.

- **Industry (Fama–French 12).** The spread is indistinguishable across industries: the group estimates run from 0.98% % (Chems) to 3.84% % (Money) across 12 groups ($Q = 4.3$, $p = 0.959$). One drift model is therefore used for the whole cross-section.
- **Size.** The heterogeneity is significant (Cochran $Q = 24.5$ on 4 d.f., $p < 0.001$). The spread falls monotonically with size, from 6.71% % ($t = 5.43$) in the smallest quintile to 1.60% % ($t = 2.41$) in the largest. This is the prediction of Section 2: the drift is greatest where arbitrage is most expensive. Size is a first-order determinant of the signal's value, so a strategy must neutralize it, not simply load on it.
- **Era.** The estimates run from 4.92% % (1996–2000) to 1.71% % (2010–2013). The across-era difference is statistically detectable (Cochran $Q = 10.8$ on 4 d.f., $p = 0.028$, independence approximation).

### 4.5 Rival hypotheses that were tested and rejected

**Surprise reversal.** Bernard and Thomas (1990) report a lag-four-quarter SUE autocorrelation of about −0.24. On 141,406 firm pairs we estimate +0.05. Firms in the top decile four quarters earlier still average a standardized surprise of +1.44 against −0.58 for the bottom decile. The data show persistence, not reversal, so no book fades a firm's own past surprise.

**Announcement-return (EAR) augmentation.** Blending the day-0 return into the signal leaves the Sharpe ratio flat (1.0–1.06 across blend weights, as recorded in `docs/HISTORY.md`). That holds once the EAR input is restricted to information realized before the first accrual day. An earlier specification used the day-0-to-day-1 return, which overlaps a position's first day of P&L, and it

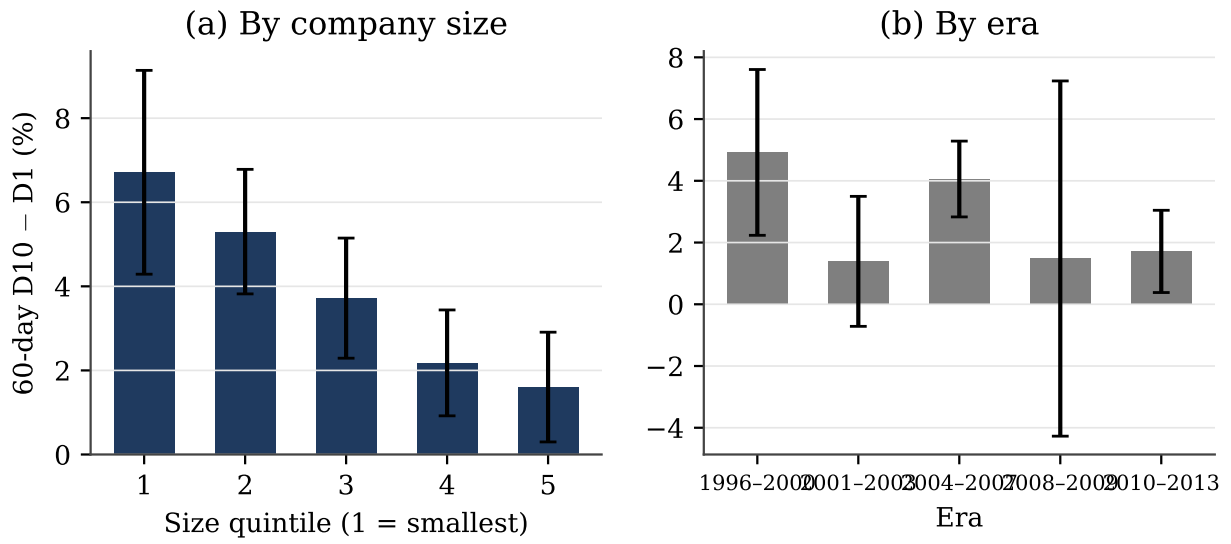



Figure 2: 60-day D10−D1 spread with 95% confidence intervals by (a) size quintile at announcement and (b) era. Source: `data/subsample_size.csv`, `data/subsample_era.csv`.

Table 4: The 60-day spread by size quintile and by era.

| Group | 60d D10−D1 (%) | $t$ | Events |
|---|---|---|---|
| Size quintile 1.0 | 6.71 | 5.43 | 21,069 |
| Size quintile 2.0 | 5.30 | 7.01 | 33,938 |
| Size quintile 3.0 | 3.72 | 5.10 | 38,766 |
| Size quintile 4.0 | 2.18 | 3.39 | 42,343 |
| Size quintile 5.0 | 1.60 | 2.41 | 46,474 |
| Era 1996–2000 | 4.92 | 3.59 | 46,600 |
| Era 2001–2003 | 1.39 | 1.29 | 26,609 |
| Era 2004–2007 | 4.06 | 6.48 | 42,457 |
| Era 2008–2009 | 1.48 | 0.50 | 22,532 |
| Era 2010–2013 | 1.71 | 2.52 | 44,513 |

produced a spurious Sharpe ratio that kept rising with the blend weight (Section 10). The machine-readable output of this diagnostic is not kept in `results/`, so these two statements rest on the project history, not on a file the build reads.

**Filtering by volatility.** Volatility does not change average returns; it changes how much a given surprise is worth (Section 6.5). Dropping whole volatility groups did not help (Section 6.5); the negative result is kept as `volatility_filter` and `calm_weighted_filter`.

### 4.6 Confirmation in a second market

If PEAD is a real mispricing of the stock, it must appear in the price of a leveraged claim on that stock. For each event with listed options we take, on the announcement day, the call (put) with delta nearest $+0.5$ ($-0.5$) and at least 95 calendar days to expiry. We hold it at its mid price to each horizon. The call spread is $+14.38$ % of premium at 60 days ($t = 4.69$) and the put spread is $-10.79$

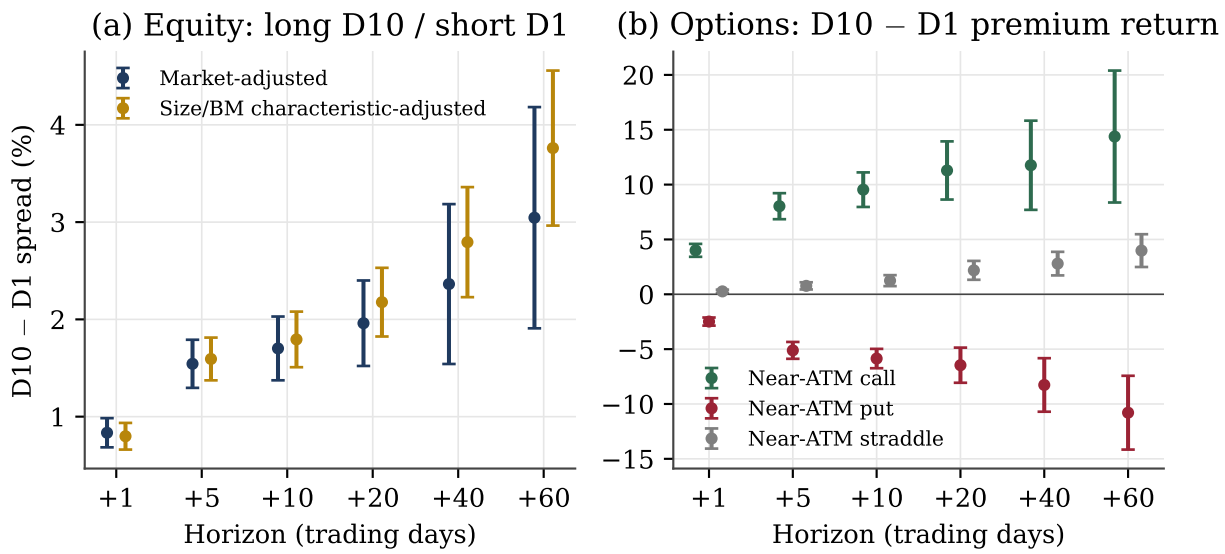



Figure 3: D10−D1 spreads with 95% quarter-clustered confidence intervals. (a) Equity, market-adjusted and characteristic-adjusted. (b) Near-the-money options, as return on premium.

% ($t$ =-6.29), the mirror image (Table 3, Figure 3).

**Magnitude check.** The ratio of the 60-day call spread to the stock spread is $14.38/3.05 \approx 4.7$. This is below the lower end of the elasticity band $\Omega \in [4.8, 6.2]$ from equation (4); part of the gap is the option's time decay over 60 days. The options market carries the same directional signal scaled by leverage.

**Direction versus volatility.** A long straddle has roughly zero delta and isolates the non-directional component. Its 60-day spread is +3.98 % ($t = 5.22$), a fraction 0.28 of the call spread. Extreme-surprise names carry some excess volatility, but the dominant effect is directional.

### 4.7 When to stop holding: the decay of the drift

A trading rule must decide when the marginal day of holding stops paying. Let $a(H)$ be the expected abnormal return from holding a position for $H$ days, net of the entry and exit cost $k$. When capital is recycled into the next opportunity, the quantity to maximize is the average net return per capital-day,

$$\mathcal{Y}(H) = \frac{a(H) - k}{H}, \qquad a(H) = \lambda|z|\big[\phi(H) - \phi(0)\big]. \tag{8}$$

The first-order condition is $a'(H^*) = \mathcal{Y}(H^*)$: hold until the marginal daily drift falls to the average daily net yield. Because $\phi$ is concave, $a'$ is decreasing, and $H^*$ is unique whenever the total drift exceeds the cost (Appendix C). Higher trading costs lengthen $H^*$.

The empirical analogue estimates a holding window for each cell of (decile × size quintile × book-to-market tercile). Let $\overline{\mathrm{CAR}}_c(d)$ be the mean characteristic-adjusted cumulative return of cell $c$ on holding day $d$. A trailing windowed slope $\varsigma_c(d) = \big[\overline{\mathrm{CAR}}_c(d) - \overline{\mathrm{CAR}}_c(d - W)\big]/W$ with $W = 20$ is compared with a cell-specific threshold of 15% of its own early peak, $\theta_c$. The decay day $H_c$ is the first day after a 30-day burn-in on which $|\varsigma_c| < \theta_c$ or the slope has changed sign, and stays

that way for 10 consecutive days. A cell with too little history backs off to (decile × size), then to the decile alone; 138 of the 144 estimated cells are the finest cells. Across cells $H_c$ ranges from 31 to 200 trading days (median 49, mean 57); the mean is 63 for D1 and 78 for D10, against 42 for D6 near the median. The windows are fitted to the whole sample, so they use information a live trader would not have; this is the third identified defect in Section 10.

# 5 The equity strategy family

## 5.1 From signal to portfolio

A book goes long positive surprises and short negative ones, with size increasing in the strength of the surprise, and it holds each position for the decay time of Section 4.7. Write $\delta_i = u_i - \frac{1}{2} \in (-\frac{1}{2}, \frac{1}{2}]$ for the centered within-quarter rank. All the neutralized books share one weight construction.

**Definition 1** (Trimmed, tilted, neutralized weights)**.** For a trim $\tau$, a tilt $\gamma$ and a per-event scale $s_i > 0$ (identically one unless a book sizes by volatility), define the conviction $t_i = \mathbf{1}\{|\delta_i| > \tau\}\ \mathrm{sgn}(\delta_i)\,(2|\delta_i|)^{\gamma}\, s_i$. For quarter $q$ and side $\sigma \in \{+,-\}$ let $\mathcal{G}_{q\sigma}$ be the set of occupied groups, where a group is a pair (industry, size quintile), and $N_{q\sigma} = |\mathcal{G}_{q\sigma}|$. The weights are

$$\tilde{w}_i = \frac{t_i}{\sum_{j \in g(i)} |t_j|} \cdot \frac{1}{N_{q\sigma}}, \qquad w_i = \frac{\tilde{w}_i}{2\,\overline{|\tilde{w}|}}, \tag{9}$$

where $\overline{|\tilde{w}|}$ is a global normalizing constant.

Within a group the weight is proportional to conviction, and each group receives the same total weight on each side. The construction targets the unpriced risks the evidence identified: industry concentration and size, the dimension along which the drift itself varies. In the code, $\tau = 0.15$ and $\gamma = 1$ for the books built on `balanced_leverage`; $s_i = 1/\mathrm{ivol}_i$ (idiosyncratic volatility, clipped) for `calm_weighted`, and $s_i = (1.02 - \pi_i)^k$ with $\pi_i$ the volatility percentile against the previous quarter's cross-section within the stock's size quintile and $k = 2$ for `calm_rank_tilt`.

**Proposition 1** (Neutrality)**.** *For every quarter's entry cohort, before any capacity cap: (a) the long and short sides have equal gross weight, so the cohort is dollar-neutral; (b) each occupied (industry, size) group receives exactly $1/N_{q\sigma}$ of its side's gross; (c) net exposure to industry k equals $\left(n_k^{+}/N_{q+} - n_k^{-}/N_{q-}\right)$ times the global constant, where $n_k^{\sigma}$ is the number of occupied size buckets of industry k on side $\sigma$. It vanishes whenever both sides occupy the same groups, and the same holds for size.*

## 5.2 Engine, sizing and costs

One engine runs every equity book. Position $i$ has dollar notional $n_i = w_i\, b\, V_q$, with $V_q$ the previous quarter-end NAV (so profits affect sizing only from the next quarter) and $b$ the base-unit fraction. The notional is capped at a fraction (5% of the stock's 21-day average dollar volume), and the gross notional entered in a quarter is capped at a multiple of trailing NAV, trimming the lowest-conviction positions first. Round-trip costs are 25/15/10/7/5 basis points by liquidity quintile (thinnest to most liquid), charged at entry and at exit. Because returns are measured against the market, each stock position is economically paired with an offsetting index position, and the daily P&L is

$$\Pi_t = \sum_{i \text{ open at } t} n_i\,(r_{i,t} - r_{m,t}) \;-\; \sum_{i \text{ entering or exiting at } t} |n_i|\,k_i, \tag{10}$$

with $k_i$ the liquidity-dependent cost. The exceptions are `market_exposed`, which uses raw returns on purpose, and the books with a market overlay (`market_boost`), which add a separate market position. Combining (1) with the weights, the expected P&L of one cohort is $\mathbb{E}[\Pi] = \lambda b V \sum_i w_i z_i [\phi(H_i) - \phi(0)]$ − costs, the covariance between conviction and drift. Since the book is balanced across industries and size by Proposition 1, the sum does not rely on any industry or size effect, only on the within-group ordering of surprises.

Two features of the engine matter for how the results are read. First, the gross cap applies to each quarter's *entry cohort*, not to the whole open book. Positions from earlier quarters are still open, so realized gross exposure can exceed the nominal cap: for `calm_weighted`, whose cap is 2.5 ×, realized gross exposure divided by NAV averages 1.8 ×, exceeds the cap on 4.0% of days and peaks at 2.8 ×. This is the second identified defect in Section 10. Second, summary statistics in `results/` are computed over the full calendar, including the idle period before the first position (the equity books start trading in 1996-01-03; the options books in 1998-01-06). Active-period Sharpe ratios are within a few hundredths of the stored ones (for example 1.51 against 1.47 for `calm_weighted`), and Table 18 uses active-period returns.

**Breadth.** The book opens about 6,449 positions a year. The fundamental law of active management (Grinold, 1989), IR $\approx$ IC$\sqrt{\text{BR}}$, with BR the number of independent bets, then implies a per-bet information coefficient of about 0.018 if every position were an independent bet (they are not, so the implied coefficient is a lower bound on the true one). The equity book is a breadth strategy: many weak bets rather than a few strong ones. Its volatility comes from the residual common factors that neutrality does not remove.

### 5.3 The family

Table 5 lists the family in lineage order, with each book's plain description from its own metadata. The progression follows the diagnoses recorded in the repository: `extreme_picks` is the textbook trade; `graded_bets` and `size_balanced` change the weights; `conviction_tilt` trims the middle of the ranking, tilts toward the extremes and drops the lowest-conviction positions first when a cap binds; `sector_balanced` adds industry neutrality; `balanced_leverage` lifts the gross cap; `market_boost` and `market_exposed` add market exposure deliberately (the first) or by accident (the second); the `calm_*` books size by volatility.

### 5.4 Headline results

**The validated book.** `calm_weighted` divides each bet by the stock's idiosyncratic volatility. It has no fitted parameter beyond the construction shared with `balanced_leverage`. It earns 6.71% a year at 4.49% volatility (Sharpe 1.47, maximum drawdown −15.6%, Calmar 0.43), compared with 6.49% and a Sharpe ratio of 1.30 for `balanced_leverage`. It has a higher Sharpe ratio than `balanced_leverage` in 3 of the three sub-periods (2.40, 1.63, 0.59 against 2.07, 1.52, 0.47). Its return is below that of the CRSP market (8.5% a year at 20% volatility, Sharpe 0.51, worst drawdown −56%, over 1996–2013): the book is market-neutral, and is not designed to beat the index.

**The research book.** `calm_rank_tilt` weights each event by $(1.02 - \pi_i)^k$, which all but switches off the most volatile stocks. Its Sharpe ratio rises along the lineage `balanced_leverage` (1.30), `calm_weighted` (1.47), `calm_rank_tilt` (1.81), and it is higher than `calm_weighted`'s in 3 of three sub-periods. **It is a research book, not a validated one:** the exponent $k = 2$ was chosen after seeing `calm_weighted`'s full-sample results, so its Sharpe ratio is in-sample. The only out-of-sample

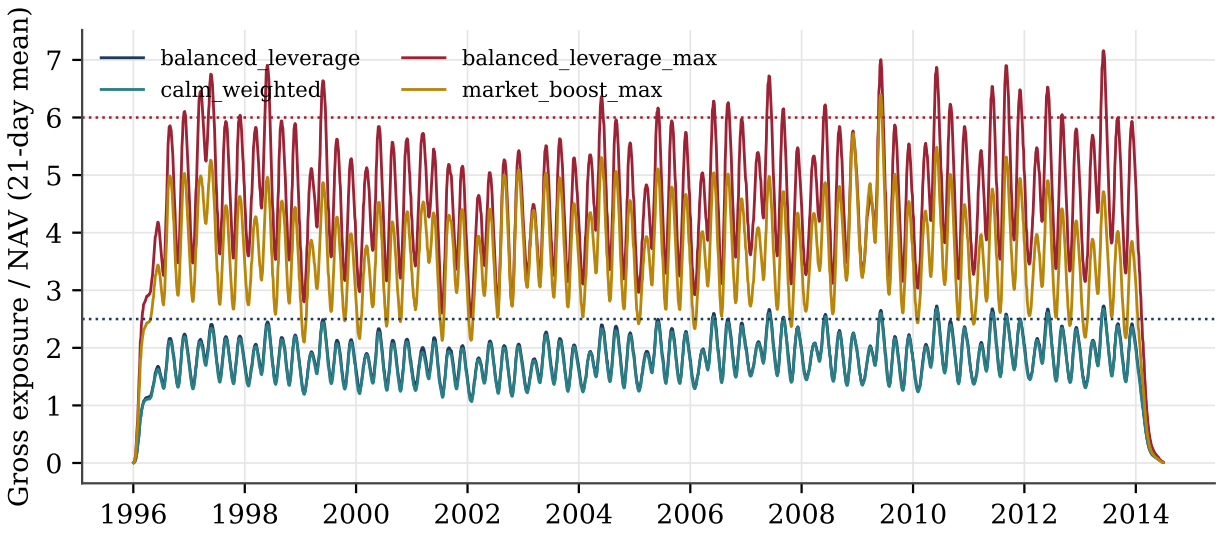



Figure 4: Realized gross exposure divided by NAV (21-day mean). Dotted lines mark the nominal caps per entry cohort. The open book exceeds its cap because earlier cohorts are still held.

evidence is a split-sample check recorded with the strategy: an exponent chosen on 1996–2004 data alone (best $k = 3$) lifts the 2005–2013 Sharpe ratio from 0.93 to 1.10, the same as the full-sample best. That is one check on one split. It supports the direction of the tilt, not the level of the Sharpe ratio.

**The reference and the cautionary books.** `balanced_leverage` is the pure-alpha reference. `market_exposed` uses raw returns and lets net exposure float; its net dollar exposure is only 10% of gross yet its beta against the market is 0.18 (alpha 5.8%), because equal dollars long and short are not equal beta. `market_boost` adds a market position of 0.5 times NAV to `balanced_leverage`: return 10.08%, volatility 10.51%, beta 0.51. It earns the market's premium on top of the alpha and carries the market's drawdown (−23.2%).

### 5.5 The leveraged and market-exposed research variants

Earlier drafts headlined an equity book earning 12.15% at 12.20% volatility (Sharpe 1.00, drawdown −36.5%). That book is the six-times-cap research variant `balanced_leverage_max`, which is identical to `sector_balanced_max`; it is not the validated headline. Each `*_max` book keeps its parent's signal and neutralization and changes only two dials, the base unit and a gross cap of 6 ×. The base unit was chosen by sweeping the unit and taking the highest return subject to a Sharpe ratio of at least 0.6 and a drawdown no worse than −40% (Section 9): an in-sample choice. Table 7 reports them with the leverage they actually use. Realized gross exposure for `balanced_leverage_max` averages 4.5 × NAV and peaks at 7.6 ×, which is prime-brokerage-level leverage, and 40.6% of its positions sit at the volume cap. Its return is higher and its Sharpe ratio is lower than `calm_weighted`'s, with a deeper drawdown. `market_boost_max` adds the market overlay to this book: it has the highest return of any book in `results/` (15.01%, which is above the market's 8.5%), a beta of 0.48 and a drawdown of −37.0%. The extra return is the market's, paid for with the market's drawdown, and it should be read as a leveraged, market-exposed research variant.

# 6 Where the equity return comes from

This section decomposes the equity returns. It rests on the saved attribution files (`results/alpha_benchmark.csv`, `situations_*.csv`, `profiles_*.csv`), which split every position's profit into a part that the surprise rank predicts ("alpha") and a part that comes from holding the average announcer.

## 6.1 Alpha against the market

Daily NAV-percentage returns are regressed on the CRSP value-weighted market return (zero risk-free rate, Newey–West standard errors with five lags), alone and with an equal-weight-minus-value-weight (size) factor. Table 8 reports the results. Across the 18 market-neutral books the largest absolute beta is 0.05; the market-exposed books (`market_exposed`, `market_boost` and their `*_max` versions) carry betas near 0.51. For `calm_weighted` the alpha is 6.9% ($t = 6.1$); adding the size factor changes it to 6.9%, so the return is not a small-cap loading. In 100% of the market-neutral books the annualized alpha against the CRSP market is lower in 2008–13 than in 1996–2001.

## 6.2 Long leg versus short leg: a benchmark artifact

Against the average of *all* announcers in the quarter, the surprise signal earns about the same on the long and short legs: for the nine profiled books the long-to-short ratio runs 0.7 to 2.2. That benchmark is biased for any cut that mixes side with size, because the average announcer is dominated by small stocks, which beat the market: big stocks structurally lag it, which penalizes shorts less and longs more depending on size mix. Measured instead against the average announcer of the *same size group and quarter* (a size-matched benchmark), the long leg earns 1.6 to 3.2 times the short leg per dollar traded (median 2.0). Table 9 and Figure 5 show both. The earlier conclusion that "the shorts pay their way as well as the longs" was a benchmark artifact; the size-matched benchmark is used from here on. The short leg is also where borrow costs fall, and they are not modelled (Section 11).

## 6.3 The best trade, and the trade that never pays

We label every position with a *recipe*: side (long or short) × how easily the stock trades (thin, mid, liquid) × beta (low, mid, high) × holding period (at most 60 days, or longer), giving 36 recipes per book.

**The best trade is the same in every book.** In 8 of the nine profiled books the single most profitable recipe is a long position in a thinly traded, low-beta stock held more than 60 days (Table 10). It uses 2%–3% of the money, earns 10%–21% of the alpha at 406–648 basis points per dollar, is positive in 68%–78% of quarters, and makes \$2.8M to \$8.0M of real profit per book. It is positive in all three eras in 6 of nine books (the exceptions: `calm_weighted`, `calm_rank_tilt`, `market_exposed`, which are negative in one era).

**High-beta shorts earn nothing, and cost real dollars.** Shorting a high-beta stock uses 11%–18% of every book's money. In 9 of 9 books the alpha of high-beta shorts is statistically indistinguishable from zero ($|t| < 2$), and in 8 of 9 they lose real dollars in all three eras. Their alpha lies between $-41$ and 47 basis points per dollar, and their real profit between $-\$21.6$M and $-\$3.3$M. Shorting low- and mid-beta stocks, by contrast, earns 71–108 basis points per dollar ($t$ from 2.2 to 3.4) in every book: same side, same signal, opposite verdicts, and beta is the dividing line. Even the good

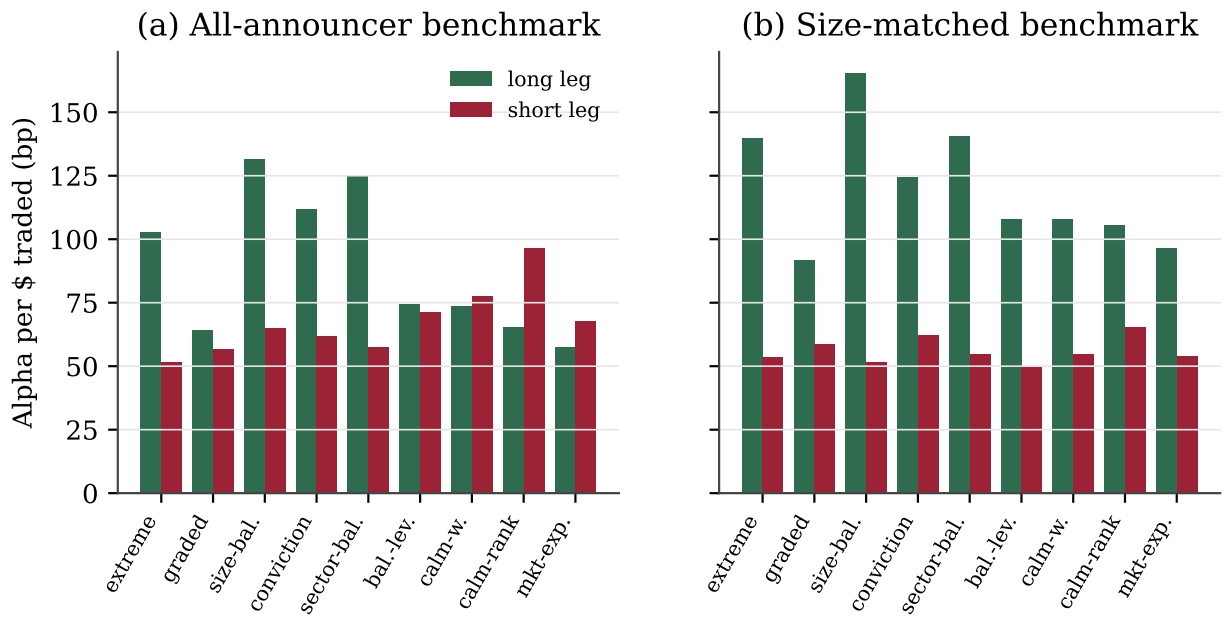



Figure 5: Alpha per dollar traded by leg under (a) the all-announcer benchmark and (b) the size-matched benchmark.

shorts lose real dollars in 9 of the nine books, because the average announcer beats the market; the long leg more than makes up for it. High-beta longs are unproven too (alpha 23–73 basis points) but profitable in real dollars ($3.7M to $23.0M), so the problem is concentrated on the short side. Whether a rule such as "no shorts in the highest-beta third" would help is untested: removing those positions also changes the hedge and the book's market exposure.

**Most of the money sits in recipes that are not individually proven.** Between 70% and 85% of each book's money is in recipes with $|t| < 2$, which still earn 40% to 54% of the alpha. Moving along the lineage from `extreme_picks` to `calm_rank_tilt` reduces the number of recipes that lose alpha from 8 (using 24% of the money) to 3 (5%). The winners and losers are drawn from the same pool: the best 1% of positions earn 1.0–2.0 times a book's total alpha, and the worst 1% lose 1.1–1.9 times it. Win rates are 51%–53%, close to coin flips, so the profit is the small net difference between large winners and large losers (Figure 7).

### 6.4 Situations: what the money looks like when cut other ways

For the reference book `balanced_leverage`, `results/situations_all.csv` tests 56 situations (company size, how easily it trades, beta, hold length, period, and others), of which 24 meet the "engine" standard ($t \geq 2.5$, positive in at least two of three eras, and above the book's average). About 56 situations are tested per book, so a few look good by chance; the three-era rule is the guard against that. Table 12 shows the cuts discussed here, and the paired-by-quarter contrasts are:

- **Hard-to-trade names pay.** Thinnest-trading versus most-heavily-traded: 266 versus 28 basis points per dollar, a difference of 238 ($t = 5.0$). Smallest versus largest company: 211 versus 31 (difference 180, $t = 2.8$). Company size and trading volume rank almost alike, so the data cannot separate them; in the joint regression neither survives once beta and hold length are

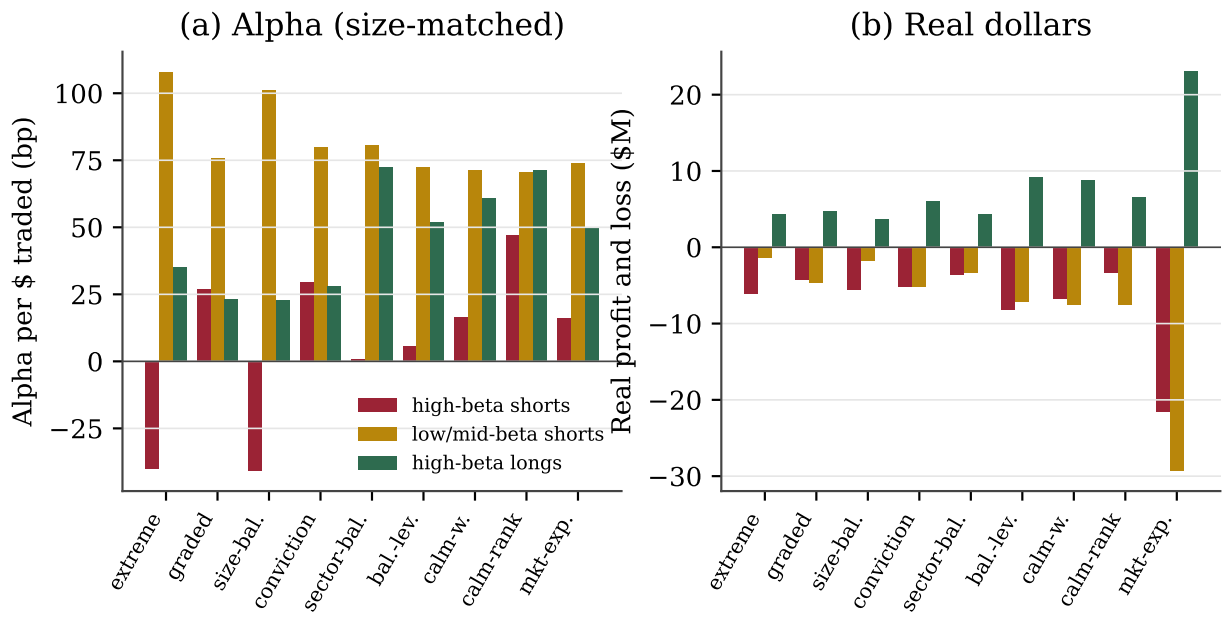



Figure 6: (a) Alpha and (b) real dollars by beta group and side, for the nine profiled books.

held fixed (size 2 basis points per standard deviation, $t = 0.1$).

- **Long holds do the work, with a caveat.** Holds over 90 days against holds up to 40 days: 315 versus 35 (difference 280, $t = 4.6$). But each cell's hold length was estimated on the same history (Section 4.7), so part of this is look-ahead.
- **Low beta pays, high beta does not.** In the joint regression one standard deviation more beta costs −39 basis points per dollar ($t$ =-3.0) with everything else fixed; longer hold windows add 55 ($t = 3.5$).
- **The edge has faded.** Alpha per dollar was 136 basis points in 1996–2001, 89 in 2002–07 and 33 in 2008–13, while the share of money traded in the last era rose to 46%. Early versus late differ by 110 ($t = 3.0$).
- **The market's direction does not matter.** Alpha per dollar is 61 when the market rose more than 5% during the hold and 43 when it fell by more than 5% (difference 6, $t = 0.0$): the market-neutral design working.

Across the nine profiled books alpha per dollar was lower in 2008–13 than in 1996–2001 in 9 of nine, by factors between 2.8 and 7.0. The mechanism section predicts such decay if arbitrage capital has grown; the data are consistent with that and do not test it.

### 6.5 Is the calm edge a volatility effect or a beta effect?

The volatility books rest on a finding about the signal: the D10−D1 spread is +4.3% % in the lowest tercile of stock-specific volatility, +3.3% % in the middle and +2.1% % in the highest ($t$-statistics 7.9, 5.1, 2.4; difference between the extremes $t = -2.1$). Volatility alone does not change average returns, so it works as a *sizing* input, not a filter; the walk-forward filters `volatility_filter` and `calm_weighted_filter` (Sharpe 1.23 and 1.41) do not beat sizing (1.47).

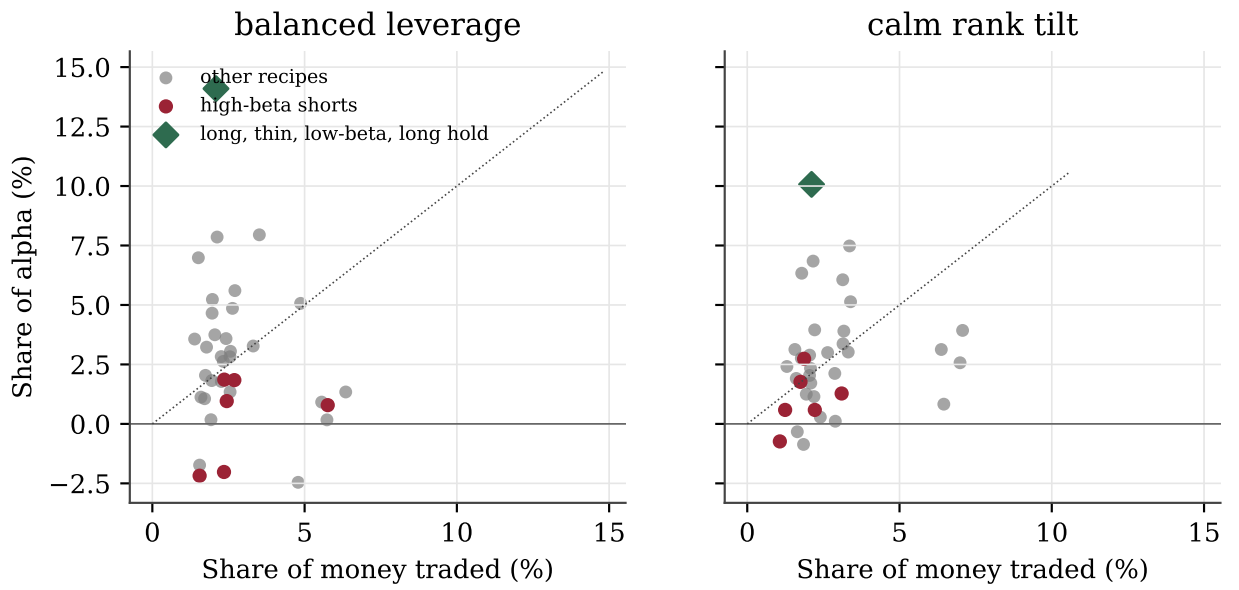



Figure 7: Each point is one of the 36 recipes: its share of the money traded against its share of alpha. Points above the dotted diagonal earn more than their share. High-beta shorts (red) cluster at or below zero; the best recipe (green diamond) earns several times its share.

Volatility and beta are correlated (correlation 0.47 between a position's beta and its volatility percentile; 51% of calm-tercile dollars are low-beta and 57% of jumpy-tercile dollars are high-beta), so the calm edge could be a beta edge. We separate them with a $3 \times 3$ cut of the `balanced_leverage` positions by beta tercile and idiosyncratic-volatility tercile (Table 13, Figure 9), using the logic of `scripts/strategy_profiles.py` on the saved positions.

Holding beta fixed, the calm-minus-jumpy gap averages 2 basis points per dollar across the three beta terciles and changes sign across them (within the low-beta tercile the jumpy positions earn 189 and the calm ones 128). Holding volatility fixed, the low-minus-high-beta gap averages 105 basis points per dollar across the three volatility terciles: 141 for calm, $-23$ for mid-volatility and 198 for jumpy stocks. The larger margin is beta. In plain terms, the calm edge is mostly a low-beta edge: most of the apparent gain from sizing by volatility is a gain from tilting toward low-beta stocks, which calm stocks tend to be. The joint regression of Section 6.4 agrees (volatility 2 basis points per standard deviation with beta, size, liquidity and hold fixed, $t = 0.1$). This does not make `calm_weighted` a worse book. It means its Sharpe gain should not be attributed to volatility itself, and that a direct beta rule is the more natural next candidate. The same cut under the all-announcer benchmark is in `docs/paper/generated/tab_betavol_all.tex` and gives the same ordering of the two margins.

### 6.6 Selecting the sizing: what was tried

The volatility-sizing family was chosen from a search whose results are recorded in `docs/STRATEGIES.md` (9 experiment rows, each with several parameter values: trim, tilt exponent, dropping deciles or holding cells, a small-cap tilt, inverse-volatility powers, the rank-based tilt, and larger base units). The relevant fact for this paper is that the search was run on the full sample, so `calm_weighted`'s construction and `calm_rank_tilt`'s exponent are in-sample

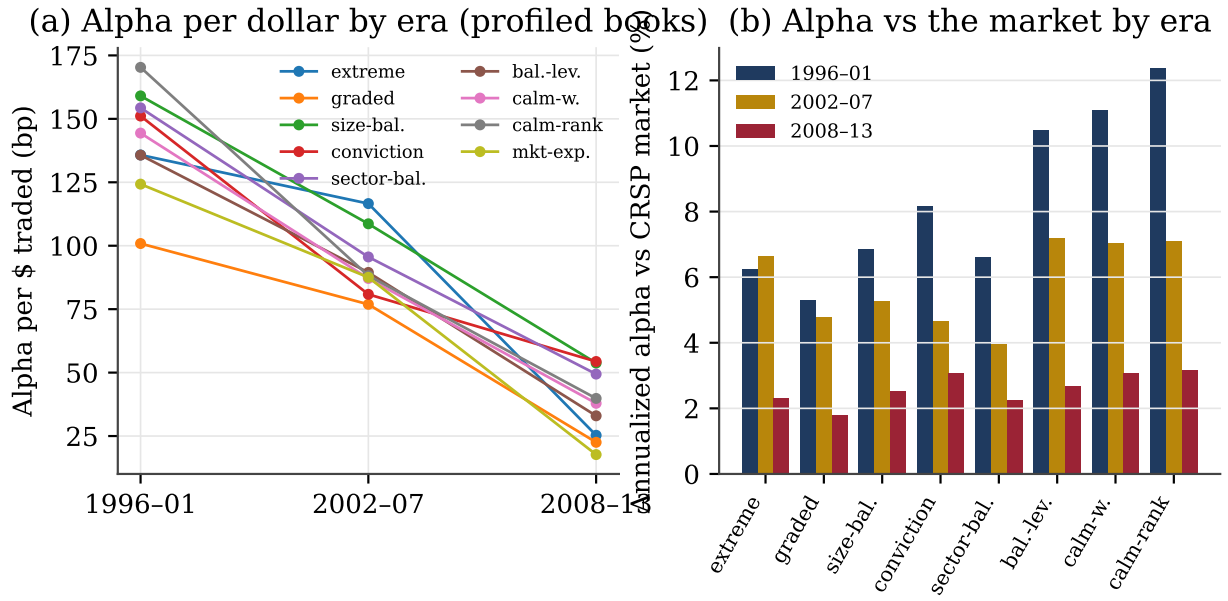



Figure 8: Decay of alpha across eras. (a) Alpha per dollar traded for the nine profiled books. (b) Annualized alpha against the CRSP market for the eight market-neutral books.

choices, with the single split-sample check noted above. Section 10.1 counts the trials that can be counted from files and deflates the Sharpe ratios accordingly.

# 7 The options strategy

The equity book earns the drift linearly. The options book earns it convexly. The question is how to choose, from the many contracts available on each event, the one that turns a modest directional edge into compounding growth, and how to size it. We use the growth-optimal (Kelly) criterion (Kelly, 1956; MacLean et al., 2011), applied contract by contract.

## 7.1 An empirical return law for each decile

For announcement quarter $q$ and surprise decile $d$, the distribution of the underlying's 60-day return is estimated empirically from *prior* quarters' events in the same decile (walk-forward, after a warm-up of 8 quarters), recentered on the trailing market drift so that the decile-specific tilt is not swamped by the unconditional equity premium. The pooled sample is cut at its quantiles into $M = 40$ equal-probability states. State $s$ has conditional mean return $\rho_{ds}$ and probability $p_s = 1/M$.

## 7.2 Payoff and objective

For an event with spot $S_0$, consider the call and put contracts $k$ of one expiry, with side $\phi_k \in \{+1, -1\}$, strike $K_k$, and premium $P_k$. The return on premium in state $s$ is

$$R_{k,s} = \frac{\left(\phi_k \left[S_0(1+\rho_{ds}) - K_k\right]\right)^+}{P_k} - 1 \ \in [-1, \infty). \tag{11}$$

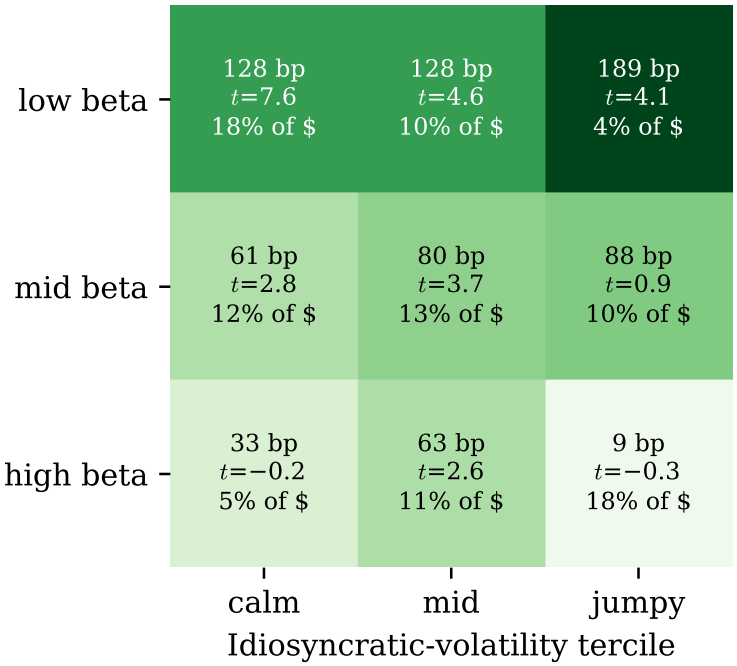



Figure 9: The $3 \times 3$ beta by volatility cut: alpha per dollar, quarter-clustered $t$-statistic and share of dollars in each cell.

Staking a fraction $f$ of capital on contract $k$ yields expected log growth

$$G_k(f) = \sum_{s=1}^{M} p_s \log\left(1 + f\, R_{k,s}\right), \qquad g_k^* = \max_{f \in \mathcal{F}} G_k(f), \quad f_k^* = \arg\max_{f \in \mathcal{F}} G_k(f), \tag{12}$$

over a grid $\mathcal{F} = \{0.05, \ldots, 0.95\}$ of 19 points. The event's contract is $k^* = \arg\max_k g_k^*$, and the event is traded only if $g_{k^*}^* > 0$. The decile-conditioned return law determines the direction on its own. In the funded book of `options_cap20`, puts are 6.5% of D1 selections and 0.2% of D10 selections (Figure 11), and calls are 98.2% of all funded trades: the selector found calls better value than puts even after bad news, which is what a return law centered on a positive trailing market drift produces.

**Proposition 2** (Well-posedness)**.** *For every $f \in \mathcal{F}$, $1 + f R_{k,s} \geq 1 - f > 0$, so $G_k$ is finite and no bet can lose more than its stake. $G_k$ is strictly concave on $[0, 1)$ unless every return is zero, with $G_k(0) = 0$ and $G_k'(0) = \mathbb{E}[R_k]$. In the continuum a positive stake is optimal if and only if the contract has positive expected return.*

**Proposition 3** (Kelly ranking is Sharpe ranking to second order)**.** *Let $\mu_k = \mathbb{E}[R_k]$, $\sigma_k^2 = \mathrm{Var}(R_k)$, and $m_k = \mathbb{E}[R_k^2] = \sigma_k^2 + \mu_k^2$. Then*

$$G_k(f) \approx f\mu_k - \tfrac{1}{2} f^2 m_k, \qquad f_k^* \approx \frac{\mu_k}{m_k}, \qquad g_k^* \approx \frac{\mu_k^2}{2m_k} = \frac{1}{2} \cdot \frac{\mathrm{SR}_k^2}{1 + \mathrm{SR}_k^2}, \qquad \mathrm{SR}_k = \frac{\mu_k}{\sigma_k}. \tag{13}$$

*Among positive-edge contracts, ranking by $g^*$ equals ranking by the single-bet Sharpe ratio. A naive selector ranks by the expected return $\mu_k$, which is unbounded in the right tail and ignores risk.*

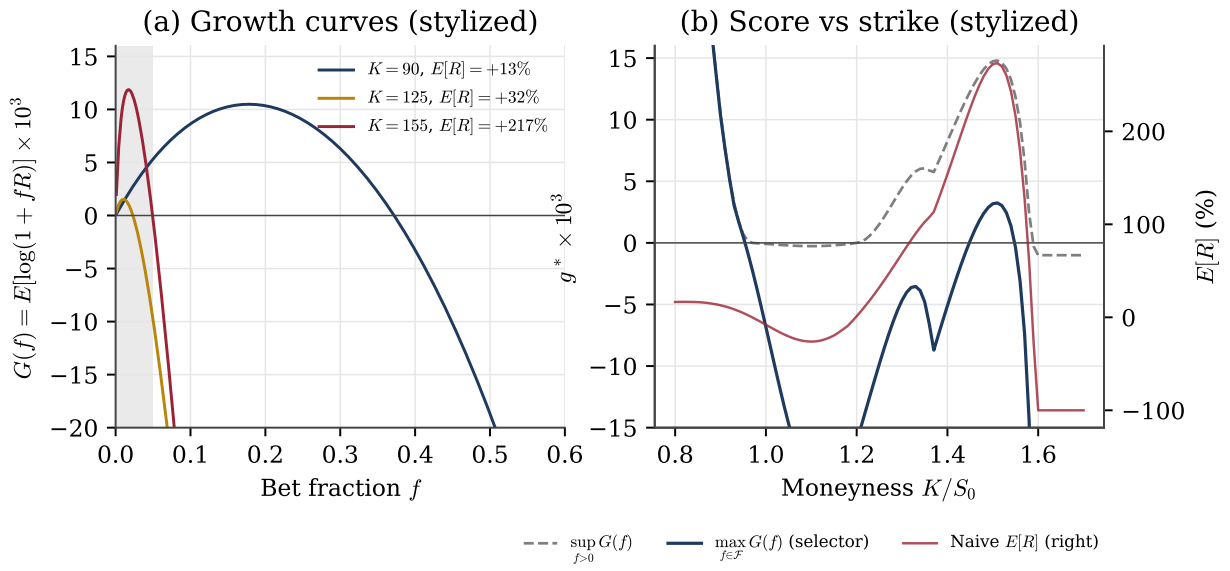



Figure 10: *Stylized illustration, not estimated from data*: a 40-state law with a diffuse body and a 4% right-tail jump, calls priced at 30% implied volatility with 120 days to expiry and valued at intrinsic at the horizon. (a) Growth curves $G(f)$: the far-out-of-the-money strike ($K = 155$) has the largest expected return and a growth peak inside the excluded region $f < f_{\min}$. (b) The continuous-stake supremum (dashed), the selector's grid score (solid) and the naive expected return across strikes.

**Proposition 4** (Minimum-stake hurdle)**.** *Since the smallest permitted stake is $f_{\min} > 0$, a contract can be selected only if $G_k(f_{\min}) > 0$, which to second order reads*

$$\mathbb{E}[R_k] \;>\; \tfrac{f_{\min}}{2}\,\mathbb{E}[R_k^2]. \tag{14}$$

The hurdle has economic content. A far-out-of-the-money strike that pays 80× its premium with probability 4% and otherwise expires worthless has $\mathbb{E}[R^2] \approx 0.04 \cdot 80^2 = 256$. With $f_{\min} = 0.05$ it must clear $\mathbb{E}[R] > \frac{1}{2} f_{\min} \cdot 256 = 6.4$, an expected return above 640%. Near-the-money contracts with moderate second moments clear it with far smaller edges. The selector's objective is thus biased toward contracts whose payoff is balanced across states, which is what a growth-optimal investor should prefer. Equation (12) rewards a strike for the *typical* payoff it delivers per unit of risk, not for the largest possible one. Figure 10 illustrates the mechanics on a stylized law; it is not estimated from data.

### 7.3 From one bet to a portfolio: the premium budget

A typical quarter offers over a thousand positive-edge events, and their single-bet fractions sum to many multiples of capital, so they cannot all be funded at $f^*$. The book instead fixes a budget: the premium outstanding across all open positions may not exceed a fraction $c$ of trailing capital. Let $O_q$ be the premium still outstanding from earlier cohorts. The quarter's budget and each event's allocation are

$$B_q = \left(c\,V_q - O_q\right)^+, \qquad \pi_i = B_q\,\frac{h_i}{\sum_{j \in \mathcal{C}_q} h_j}, \qquad h_i = 0.5\,f^*_{k^*(i)}, \tag{15}$$

so capital is allocated in proportion to each event's Kelly fraction (the factor 0.5 is the half-Kelly multiplier, which cancels in the proportion). Contracts are whole lots of 100; a round-trip cost of

3% of premium, with a floor of $0.05 per share (one option tick), is charged at entry. Positions are marked at the quoted prices on days 1, 5, 10, 20, 40, 60 and exit at day 60.

**Proposition 5** (Proportional allocation is fractional Kelly)**.** *(a) The allocation is Kelly-proportional in relative terms, $\pi_i = \lambda_q f_i^*$ with $\lambda_q = B_q / \sum_j f_j^*$; its absolute scale is set only by the budget c. (b) Under independence and the quadratic approximation of Proposition 3, scaling the vector of single-bet Kelly stakes by $\lambda \in (0, 1]$ gives portfolio growth $(2\lambda - \lambda^2) \sum_i g_i^*$ and a portfolio Sharpe ratio*

$$\mathrm{SR}_{port} = \frac{\sum_i \mu_i^2 / m_i}{\left( \sum_i \mu_i^2 \sigma_i^2 / m_i^2 \right)^{1/2}}, \tag{16}$$

*which does not depend on $\lambda$ and grows as $\sqrt{N}$ in the number of bets. (c) Every position's loss is bounded by its premium, so the loss on all open positions is at most c times the capital at which they were sized.*

Part (b) says that the book's Sharpe ratio comes from *breadth*, not from leverage: `options_cap20` funds about 4,433 small bets a year, an implied per-bet information coefficient of 0.027 (a lower bound on the true value, for the reason given for the equity book). It also predicts that return and volatility scale roughly linearly in the budget, with the Sharpe ratio nearly unchanged: raising the cap from 20% to 30% moves the return from 4.12% to 5.69% and the volatility from 2.30% to 3.61%, while the Sharpe ratio goes from 1.77 to 1.55. The cap is a risk-appetite dial with a hard loss interpretation (part (c); the largest outstanding premium in the `options_cap20` run was 18.8% of NAV), not a source of extra edge. The caps 20% and 30% are in-sample choices: they were fixed after the earlier results, and larger caps (40%) exist only in the legacy runs.

### 7.4 Corrected results

Table 14 reports the books. The validated headline is `options_cap20`: 4.12% a year at 2.30% volatility, a Sharpe ratio of 1.77 and a maximum drawdown of −13.8%. The 30% book `options_cap30` returns 5.69% at a Sharpe ratio of 1.55. The pessimistic bound `options_worst_case`, in which a dying option is worth nothing, returns 2.34% at 0.99. The funded book has 65,003 trades.

**Where the return comes from.** 98.2% of funded trades are calls, producing 96.5% of gross profit. The pay-off is convex: the win rate is 37%, the median trade returns −24% of premium, and the premium-weighted mean return is 25%; the best 1% of trades earn 101% of the net profit, which is more than the whole net profit. Profit is concentrated in time: 2003 and 2013 contribute 62% of net profit, and entry years 1999, 2001, 2002, 2008 lose (Figure 11). The edge is a convex bet on the upside, harvested across thousands of small positions; with puts so rare among the funded trades it is not a symmetric drift trade.

### 7.5 The quote-marking error

The original backtest valued each position at days 1, 5, 10, 20, 40, 60 from the contract's mid quote and, when a quote was missing, *carried the previous quote forward*. 22% of picks had no mid quote at 60 days, and re-querying OptionMetrics showed that 93% of those were not missing: the bid was zero, so the option had died and nobody would buy it. Carrying its last live price forward never marks a dying option down, which flatters exactly the contracts that die. The correction marks a zero-bid checkpoint at half the ask (or zero in the pessimistic bound) and adds a floor of one option tick of round-trip cost per share. For the 20% book the effect is: 8.58% and a Sharpe ratio of 3.78 as originally reported; 5.87% and 2.50 with the corrected mark; 4.12% and 1.77 with the cost floor

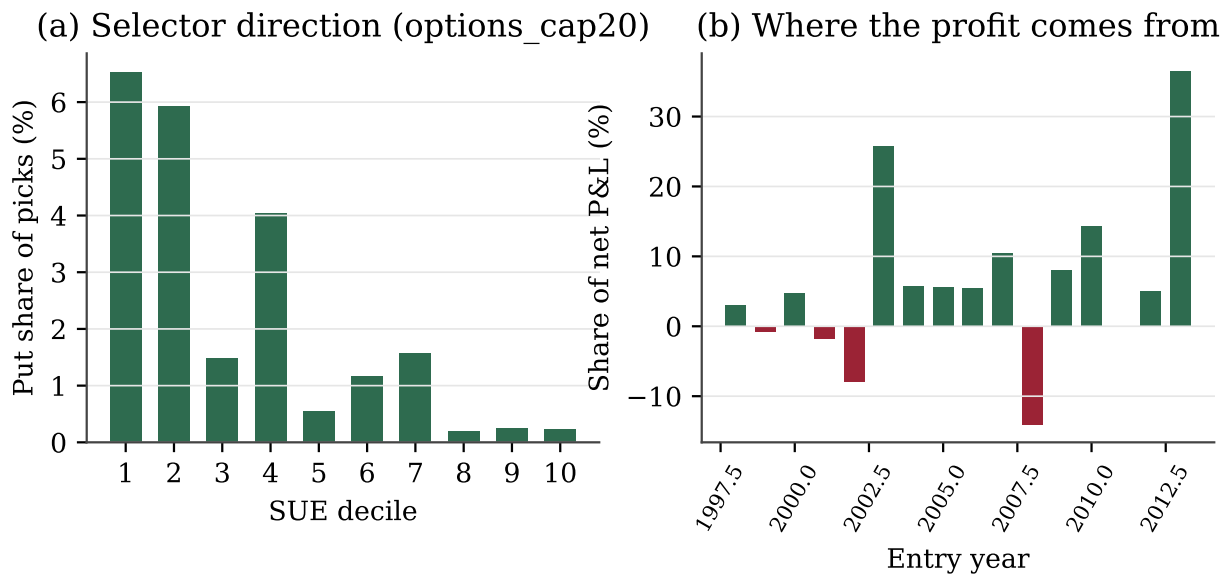



Figure 11: Funded `options_cap20` trades. (a) Share of puts among the selector's picks by SUE decile. (b) Share of net profit by entry year.

(`options_cap20`); and 2.34% and 0.99 if dead options are worth zero (`options_worst_case`). For the 30% book the legacy figures are 12.26% and a Sharpe ratio of 3.49, against 5.69% and 1.55 after the correction. The legacy convention overstated the return by a factor of 2.1 and the Sharpe ratio by a factor of 2.1 for the 20% book. Figure 12 shows the same trades under the three conventions. The earlier drafts' options headline (12.26% and a Sharpe ratio of 3.49, the 30% legacy book) is therefore an overstated number and appears only here and in the audit log.

**Remaining weaknesses, unquantified.** The cost floor is a minimum, not a spread model; contracts with premiums of a few tenths of a dollar (median entry premium $1.19 across funded trades) can have quoted spreads far wider than one tick. There is no market impact, although the book opens about 4,433 trades a year. The first walk-forward distributions rest on 8 quarters. Profit is concentrated in few years.

# 8 The combined strategy

## 8.1 Why the two books combine

Let the two sleeves have annualized volatilities $\sigma_e, \sigma_o$, correlation $\rho$, and sleeve weights $w_e, w_o$ (multiples of each sleeve's own sizing, which need not sum to one). The blend's volatility is

$$\sigma_p^2 = w_e^2\sigma_e^2 + w_o^2\sigma_o^2 + 2w_e w_o \rho\, \sigma_e\sigma_o. \tag{17}$$

Over the options book's active window the daily correlation between `calm_weighted` and `options_cap20` is $-0.05$ (monthly: $-0.16$ over 175 months); the equity sleeve's correlation with the market is $-0.06$ and the options sleeve's is 0.20. The cross term is therefore negligible or negative. The correlation is low for an economic reason. Both books earn from the same events, but the equity book's short leg loses in a rally led by low-quality, heavily shorted stocks, the regime that produces

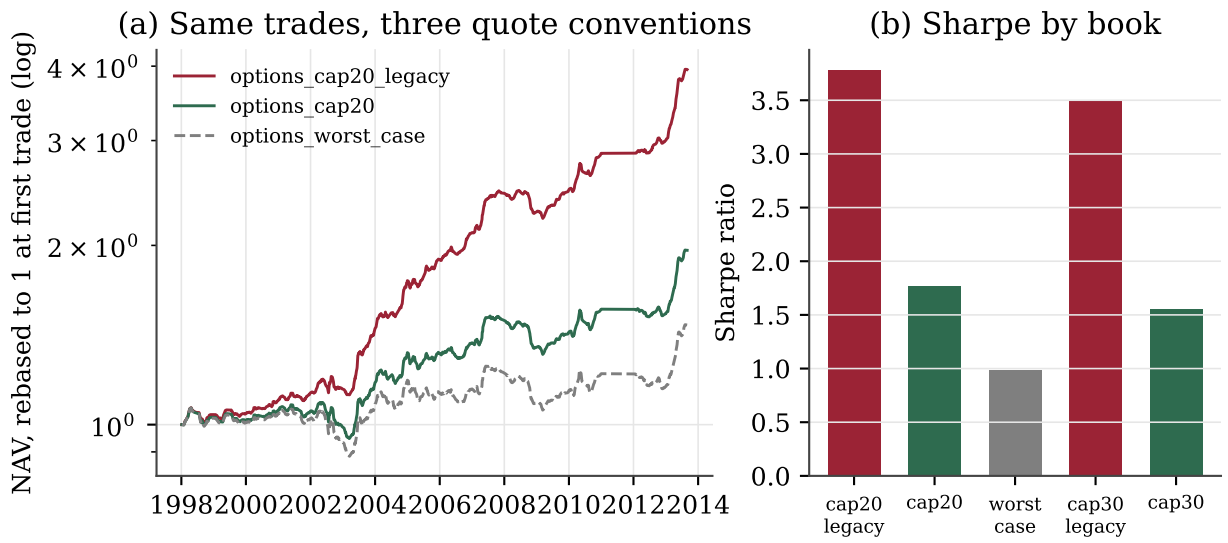



Figure 12: (a) NAV of the same Kelly trades under the legacy quote convention (carry-forward), the corrected mark and the pessimistic bound. (b) Sharpe ratios by book; red marks the overstated legacy books.

momentum crashes (Daniel and Moskowitz, 2016), and it has no cap on that loss. A purchased put loses at most its premium, and calls on positive-surprise names participate in the rally.

### 8.2 The blends

The blends are `hedged_blend` $= 0.8 \times$ `calm_weighted` $+ 1.6 \times$ `options_cap20` on one NAV rebalanced daily (the validated blend; the multiples give each sleeve about equal stand-alone volatility), and `calm_blend` $= 0.9 \times$ `calm_rank_tilt` $+ 1.6 \times$ `options_cap20` (the research blend, inheriting the in-sample status of its equity sleeve). `legacy_blend` uses the overstated options sleeve and is an upper bound, not a result. The blend multiples are chosen so that each sleeve contributes about equal risk, which is an in-sample choice; the weights are multiples of each sleeve's own sizing, not shares of capital, because the options sleeve spends only its premium budget and the equity sleeve is margin-financed long/short.

**Results.** `hedged_blend` earns 11.33% a year at 4.83% volatility with a Sharpe ratio of 2.25 and a maximum drawdown of −16.7%; its Sharpe ratio exceeds the Sharpe ratio of either sleeve alone (1.47 and 1.77), and it is above the weaker sleeve's Sharpe ratio in every era. The sleeves' era Sharpe ratios are 2.40, 1.63, 0.59 (equity) and 0.69, 2.04, 2.39 (options): the equity sleeve has the higher Sharpe ratio in the first era and the options sleeve in the last. The blend's worst era has a Sharpe ratio of 1.91, against 0.59 and 0.69 for the sleeves. The blend's beta against the market is 0.033, so a market hedge would add cost without removing exposure. `calm_blend` earns 12.43% at 4.73% volatility (Sharpe 2.50, drawdown −15.5%). `legacy_blend`'s Sharpe ratio is 1.43 times `hedged_blend`'s; that gap is the effect of the quote-marking error on the blend.

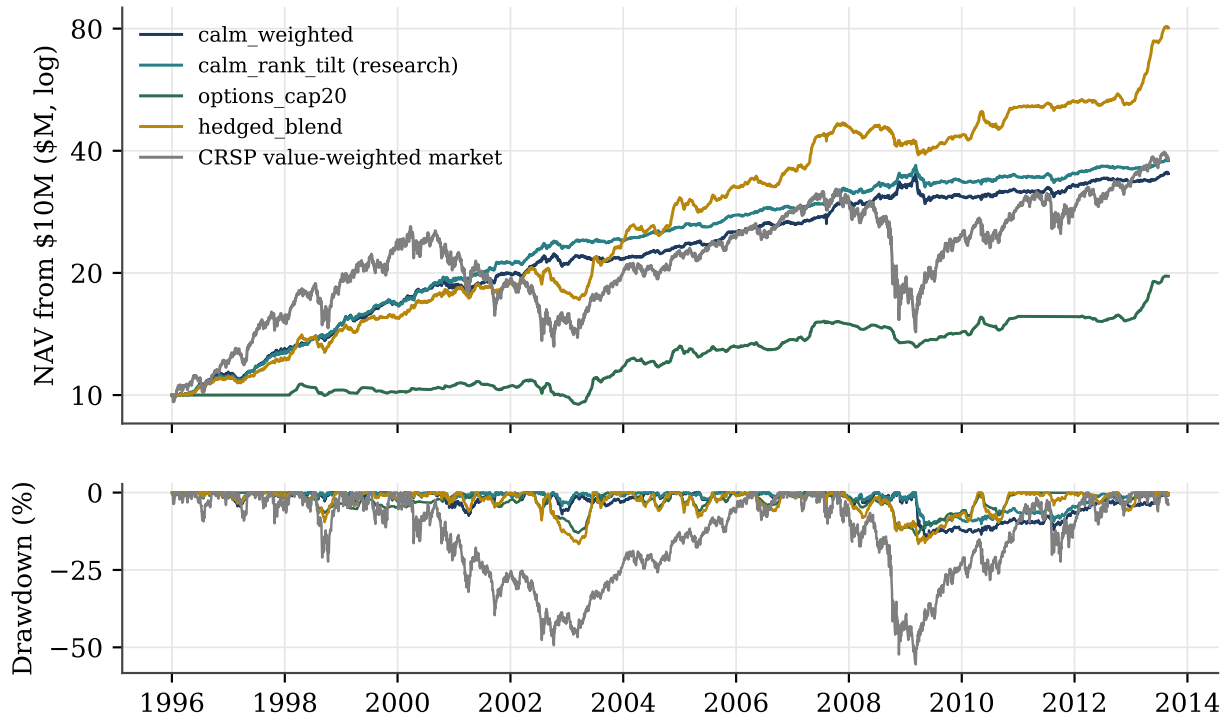



Figure 13: NAV (log scale, $10M start) and drawdowns of the validated equity book, the research equity book, the corrected options book, the validated blend and the CRSP value-weighted market, 1996 to August 2013.

### 8.3 Calendar years

The equity book has 1 losing calendar year(s) in 1996–2013 (2009); the worst was −6.9% in 2009, when the options book returned 6.1%; in 2008 the equity book returned 10.6% and the options book −10.0% (Table 16, Figure 14). Two years are thin evidence of diversification: the 2009 rebound was a rally in low-quality, heavily shorted names, the regime the equity short leg is exposed to.

## 9 Capacity and the risk-budget dial

**Proposition 6** (Concave capacity frontier)**.** *Let $b$ scale the book, and assume expected abnormal return per dollar, $\alpha_i$, is non-increasing in a stock's liquidity. Then the marginal expected alpha per dollar of additional capital is non-increasing in $b$. Holding the volatility cost fixed, expected arithmetic return is concave in $b$, and the geometric return (arithmetic return less $\frac{1}{2}\sigma^2(b)$) has an interior maximum.*

Economically, as capital grows, the stocks with the highest drift saturate their liquidity limits first. Additional capital flows to larger, lower-drift names, while risk rises in proportion to gross exposure. The return of the `balanced_leverage` construction at a six-times cap, swept over the base unit $b$ (Table 17, Figure 15a), peaks at the chosen base unit: it peaks at $b = 12 \times 10^{-3}$ with a return of 12.15%, where 27.9% of positions are volume-capped, and falls to 0.39% at $b = 36 \times 10^{-3}$ with 63% capped. The rule used to choose $b$ for the `*_max` books (highest return subject to a Sharpe ratio of at least 0.6 and a drawdown no worse than −40%) selects $b = 12 \times 10^{-3}$ from this grid, in sample. The sweep row at the chosen base unit reproduces the saved `balanced_leverage_max` run.

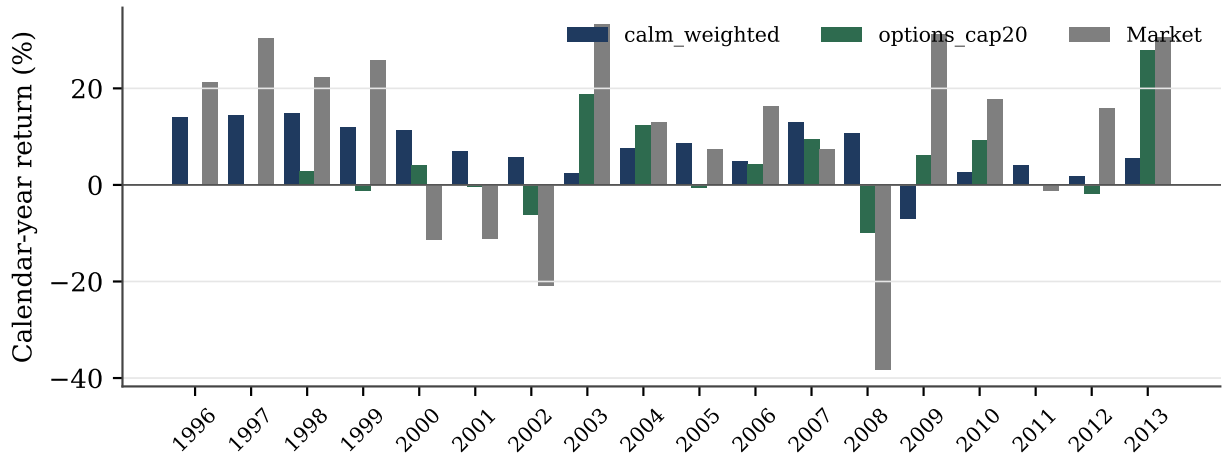



Figure 14: Calendar-year returns of the equity book, the options book and the CRSP value-weighted market.

Equity return is capacity-bound; options return is budget-bound. For options, to first order return and volatility scale with the premium cap $c$, which is a risk-appetite dial (Figure 15b). The practical limits on the options book lie outside the backtest: quoted width, depth and market impact in single-name options. Among capital-split blends (sleeve weights summing to one; panel c), the Sharpe ratio runs from 1.47 to 2.25 and peaks at 30% equity (return 4.5%, volatility 2.0%), a lower-risk and lower-return corner than the multiples used for `hedged_blend`.

## 10 Statistical robustness and backtest hygiene

### 10.1 Multiple testing

Both books were selected from a family of trials, so we deflate their Sharpe ratios (Bailey and López de Prado, 2014). For $N$ trials whose annualized Sharpe ratios have variance $V$, the expected maximum under the null is

$$\widehat{\mathrm{SR}}_0 = \sqrt{V}\left[(1-\gamma_E)\,\Phi^{-1}\!\left(1-\tfrac{1}{N}\right) + \gamma_E\,\Phi^{-1}\!\left(1-\tfrac{1}{Ne}\right)\right], \tag{18}$$

where $\gamma_E$ is the Euler–Mascheroni constant. The deflated Sharpe ratio is the probabilistic Sharpe ratio (Bailey and López de Prado, 2012) measured against $\widehat{\mathrm{SR}}_0$, allowing for the skewness $\hat{\gamma}_3$ and kurtosis $\hat{\gamma}_4$ of the returns:

$$\mathrm{DSR} = \Phi\left(\frac{(\widehat{\mathrm{SR}} - \widehat{\mathrm{SR}}_0)\sqrt{T-1}}{\sqrt{1-\hat{\gamma}_3\widehat{\mathrm{SR}} + \frac{\hat{\gamma}_4-1}{4}\widehat{\mathrm{SR}}^2}}\right), \tag{19}$$

with Sharpe ratios in per-period units and $T$ the number of active days.

**How the trials were counted.** Every count is a count of entries in a committed file. For equity, $N = 326$: the 22 equity runs in `results/` (of 22 registered equity strategies) plus the 304 rows in the 13 sweep files `data/sweep_*.csv` (base unit, liquidity cap, leverage and trim sweeps for the

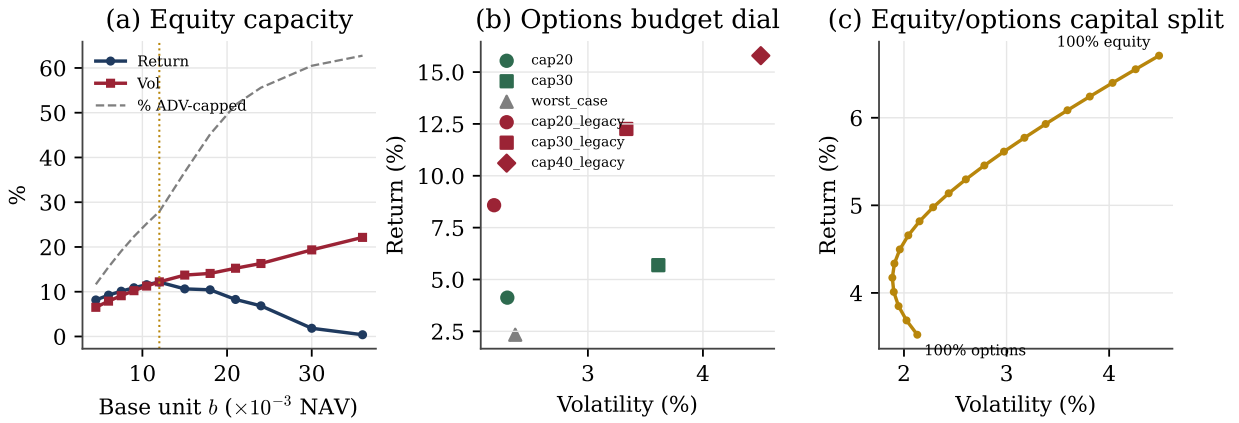



Figure 15: (a) Equity: return, volatility and share of volume-capped positions against the base unit. The dotted line marks the chosen unit. (b) Options: return and volatility of the premium-cap books (green: corrected; red: overstated legacy). (c) Capital-split frontier of the equity and options sleeves from `results/portfolio/frontier.csv` (`calm_weighted` with `options_cap20`).

numbered designs). The standard deviation of their Sharpe ratios is 0.24. For options, $N = 30$: the 6 options runs in `results/` plus the 24 rows of the 4 horizon-by-budget comparison files from before the correction (four constructions $\times$ six holding horizons); the standard deviation of their Sharpe ratios is 1.30, but across only the 3 corrected runs it is 0.40. These counts are lower bounds. The experiment tables recorded in `docs/STRATEGIES.md` (9 rows of several parameter values each) are not files with Sharpe ratios and are not counted, nor are discarded designs that left no file. Table 19 therefore shows how the answer moves with $N$.

**Results.** The validated equity book `calm_weighted` clears the 0.95 bar at the counted $N$ (DSR 1.00); its break-even is about 251,189 independent trials, against the 326 counted. The research book `calm_rank_tilt` clears the bar at the counted $N$ (DSR 1.00). The six-times-leveraged `balanced_leverage_max` has a DSR of 0.91 and a break-even of 139 trials. The options book `options_cap20` clears the 0.95 bar only when trial dispersion is measured across the corrected runs; measured across all counted trials (which include the overstated legacy runs) it does not: its DSR is 0.00 with the dispersion of all counted trials and 1.00 with the dispersion of the corrected runs only (0.00 and 1.00 for `options_cap30`). The honest reading is that the options Sharpe ratio is not distinguishable from selection when one allows the dispersion that the overstated runs displayed, and is when one does not. The deflation does not cover choices that left no trial record: the decay-day windows, the trim and tilt, the volatility-sizing family and the exponent of `calm_rank_tilt`, the options caps, and the blend multiples are all in-sample selections.

## 10.2 Audit log: defects found

Table 20 lists every defect found, with the number before and after where the saved results allow it. We report them because the *direction* of each correction shows how fragile each headline number is: every correction applied here lowered or left unchanged the reported return, except that the correction to the Sharpe denominator raised Sharpe ratios.

The first six rows can be traced to files in this repository. The remaining rows were found and fixed before the saved runs; their before/after numbers predate `results/` (and in two cases rest on

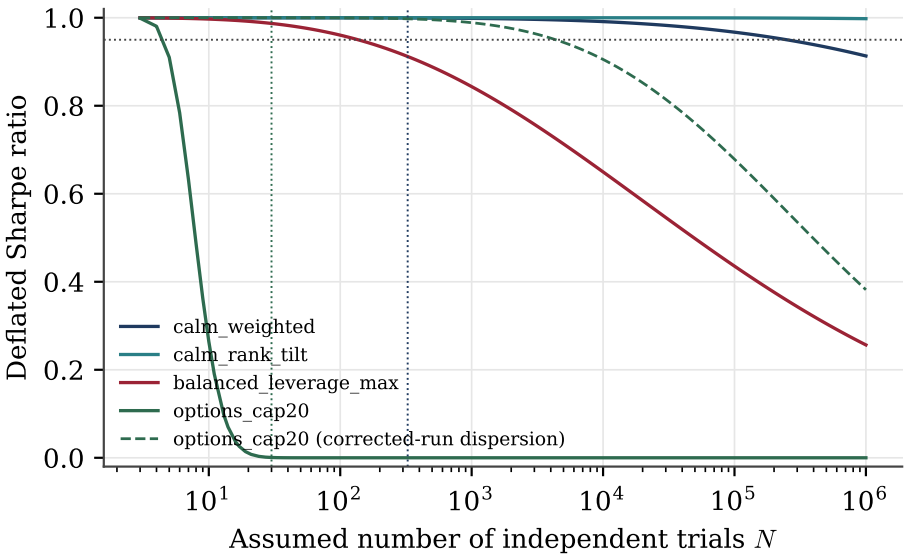



Figure 16: Deflated Sharpe ratio against the assumed number of trials. Dotted verticals mark the counted $N$ for equity and for options; the dashed green line recomputes the options curve with the dispersion of the corrected runs only.

the project history in `docs/HISTORY.md`), so we state the direction of the effect and not a number. The three rows marked *identified* are the open items: identified, not yet applied to the reported numbers (no `docs/FIXES.md` exists in this checkout). No equity book has a legacy twin in `results/` yet, so no before/after comparison of the three equity corrections can be produced from the saved runs.

For reference, the three worst days of `options_cap20` are 2002-09-13, 2008-10-24, 2010-05-21 (the worst loses $-0.8\%$ of NAV in a day); we have not matched them to market events.

## 11 Limitations and what is not modelled

We list these without softening. Each is either a reason realized performance would be lower or a test still to be run.

1. **Sample period and decay.** All results are for 1996–2013, and alpha per dollar was lower in the last era in 9 of nine profiled books (by factors between 2.8 and 7.0). A data extension and a post-publication decay adjustment (McLean and Pontiff, 2016) are required before any capital is allocated.

2. **Look-ahead we have identified but not removed.** Intra-quarter decile ranking and full-sample holding windows (Section 10.2). Both flatter the reported numbers by an unknown amount.

3. **In-sample selection.** The exponent of `calm_rank_tilt`, the volatility-sizing family, the trim and tilt, the base unit and the six-times cap of the `*_max` books, the options premium caps, the blend multiples, and the holding windows were chosen on the sample they are reported on. The split-sample check in Section 5.4 is the only out-of-sample evidence, and the deflated Sharpe ratio of Section 10.1 covers only the counted trials.

4. **Borrow fees and financing (equity).** Neither is modelled, and the short leg is small-cap-heavy. The short side averages 0.79 × NAV for `calm_weighted`, so each 1% a year of borrow cost removes 0.79% of return; a fee of 8.4% would remove all of it, and a uniform 5% would leave 2.7% (the same arithmetic for `balanced_leverage`: 2.4%). Hard-to-borrow names cost more, and the model of Section 2 predicts they are the names with the most drift. Gross exposure above NAV and the stacked blend multiples also need financing.
5. **Market impact and capacity (equity).** The 5% volume cap is a participation limit, not an impact model. 11.0% of `calm_weighted`'s positions and 40.6% of `balanced_leverage_max`'s sit at the cap (Frazzini et al., 2018; Novy-Marx and Velikov, 2016). Costs are 5–25 basis points per side by liquidity quintile.
6. **Leverage.** The gross cap applies per entry cohort; realized open-book leverage is higher than the cap (Section 5.2), and the `*_max` books run at several times NAV.
7. **Option execution.** Entries use the ask and dead quotes a half-ask mark or zero, with a flat 3% charge and a one-tick floor. Quoted spreads on out-of-the-money contracts are often wider, and the book opens about 4,433 positions a year; capacity and impact have not been modelled. Replacing the floor with quoted bid/ask on the selected contracts at entry and exit is the most important outstanding robustness test.
8. **Options data coverage.** 2011 is missing because its source file is structurally corrupt, so the trading-day calendar compresses across the gap, and 2013 ends in August. The matched sample is tilted toward names with liquid, longer-dated chains.
9. **Return-law pooling.** One distribution per (decile, quarter) is shared by every stock in the decile, regardless of its own volatility.
10. **Live universe.** The backtest trades the full CRSP event universe. A live book restricted to large stocks earns much less per dollar: for `balanced_leverage`, 31 basis points in the largest company-size quintile against 211 in the smallest.
11. **Reporting convention.** Stored summary statistics cover each book's full calendar, including idle periods before the first position; the active-period figures differ slightly (Section 5.2). The equity daily series runs to 2014-06-30 while events end in 2013.
12. **What the earlier drafts claimed that is not repeated.** The earlier drafts quoted an equity book at 12.15% as the strategy, an options book at 12.26% and a Sharpe ratio of 3.49, and described forward-filled missing quotes as a caveat rather than the error it was. None of these is a headline here.

## 12 Conclusion

Post-earnings-announcement drift in this sample passes the tests an allocator should require. It is significant under earnings-season clustering. It survives a size/value characteristic benchmark, which enlarges the spread. It is graded by firm size as limits to arbitrage predicts, and it appears in a second, independently priced market at roughly the magnitude implied by option elasticity. We harvest it two ways. The linear book neutralizes the unpriced exposures we could measure and earns 6.71% a year in its validated form (`calm_weighted`, Sharpe 1.47) with no market beta; its research counterpart `calm_rank_tilt` earns 7.12% (Sharpe 1.81), but with an exponent chosen in sample. The convex book chooses, event by event, the contract with the highest expected log growth

and earns 4.12% (Sharpe 1.77) after the quote-marking correction, not the 8.58% first reported. The two are close to uncorrelated and hedge each other's worst year; the validated blend earns 11.33% at a Sharpe ratio of 2.25.

The decomposition adds three cautions. The edge is concentrated in a kind of trade: long positions in thinly traded, low-beta stocks held more than two months. High-beta shorts earn no alpha and lose real dollars. And the calm edge is mostly a low-beta edge. The long leg earns more than the short leg once the benchmark is size-matched, alpha per dollar has fallen across the sample, and the places where the drift is largest are the places where the costs we do not model are largest. Three corrections are identified and not yet applied, and the multiple-testing adjustment covers only the trials that left files. The work that remains before deployment is concrete: a walk-forward re-estimation of the holding windows, trailing-breakpoint ranking, borrow and financing costs, quote-level option execution costs, and data after 2013.

**Disclosure.** All performance figures are hypothetical, are based on historical backtests (1996–2013) net of modeled trading costs, and do not reflect actual trading. They benefit from hindsight in model design, and the modeling assumptions in Section 11 omit costs that would lower them. Realized results may differ materially. Nothing here is an offer, a solicitation, or investment advice.

# A Names used in earlier drafts and here

# B Parameters and reproducibility

**Weights.** Trim $\tau = 0.15$ and tilt $\gamma = 1$; groups are (side, size quintile, Fama–French industry) within each announcement quarter; the weights are rescaled to a mean absolute value of 0.5.

**Holding windows.** Per-cell decay days (Section 4.7), from 31 to 200 trading days.

**Runs.** The 31 saved runs in `results/` were produced between 2026-10-01 and 2026-10-01 at git revision(s) 9044ee2 under pandas 3.0.5; results reproduce only under the pinned pandas 3 (see `requirements.txt`).

**Where every number comes from.** Per-book statistics: `results/<name>/summary.json` and `daily.csv.gz` (realized leverage, active-period Sharpe, calendar returns, correlations, deflated Sharpe ratios). Leaderboard checks: `results/leaderboard.json` and `leaderboard.md`. Alpha against the market: `results/alpha_benchmark.csv`. Situations: `results/situations_*.csv`. Recipes, leg and beta splits, timing and tails: `results/profiles_*.csv`. Capital-split frontier: `results/portfolio/frontier.csv`. Signal evidence (Sections 3–4): the committed tables under `data/` (`decile_horizon_stats`, `spread_horizon_stats`, the `ff_*` and `option_*` versions, `subsample_*`, `sue_reversal_summary.json`, `cell_decay_days.csv`, `vol_decile_diagnostic_summary.json`), plus `data/sweep_*.csv` and the options comparison files for the trial count and the capacity table. Construction constants (trim, tilt, exponent, costs, grids) are read from the strategy classes and engine modules. Three groups of numbers need files that are not committed: the $3 \times 3$ beta-by-volatility cut and the per-trade options statistics (from `results/*/positions.parquet`, git-ignored) and the CRSP market series (licensed). They are recomputed whenever the main repository's data is reachable (environment variable `PEAD_MAIN_REPO`) and the run it holds is the run in `results/`; otherwise the committed cache in `docs/paper/cache/` is used, and the build warns when the cache is stale.

**Rebuilding.** After any change to `results/` or `data/`, one command rebuilds the statistics, tables, figures and PDF with no hand edits:

```
python -m pead.reporting.paper
```

(equivalently `python -m pead paper`; `--no-pdf` skips the LaTeX step, `--refresh-external` recomputes the cached quantities). The build fails if a number macro used here is undefined, if a strategy name used here is not in `results/`, if `results/leaderboard.md` disagrees with the saved summaries, or if a numbered legacy label appears in the text.

**Code.** Strategy files: `pead/strategies/`; engines: `pead/engines/`; reports and attribution: `pead/reporting/`; this paper: `pead/reporting/paper.py` and `paper_figures.py`.

# C Proofs

**Concavity of the incorporation profile.** A Gamma$(k, \theta)$ mixture of exponential rates has Laplace transform $\mathbb{E}[e^{-\kappa d}] = (1 + d/\theta)^{-k}$, which gives (3). Then $\phi'(d) = (1 - \phi(0))\,(k/\theta)\,(1 + d/\theta)^{-k-1} > 0$ and $\phi''(d) < 0$, so $\phi$ is increasing and strictly concave with $\phi(d) \to 1$. Since $\Delta(h) \propto \phi(h) - \phi(0)$, the spread is increasing, concave and bounded. The unincorporated fraction decays as a power, $(1 + d/\theta)^{-k}$, not exponentially, so the profile has no single time scale and keeps rising long after a single-speed model would have flattened.

**Uniqueness of the holding period.** Write $\mathcal{Y}(H) = (a(H) - k)/H$ with $a$ increasing and concave, $a(0) = 0$, and $a(\infty) > k$. Then $\mathcal{Y}'(H) = N(H)/H^2$ with $N(H) = a'(H)H - a(H) + k$, and $N(H) = 0$ if and only if $a'(H) = \mathcal{Y}(H)$. Now $N'(H) = a''(H)H < 0$, $N(0) = k > 0$, and $N(H) \to k - a(\infty) < 0$ as $H \to \infty$ (because $a'(H)H \to 0$ for a concave bounded $a$). So $N$ has exactly one zero $H^*$, and $\mathcal{Y}$ increases before it and decreases after, which makes $H^*$ the unique maximizer. □

**Proof of Proposition 1.** Fix $q$ and side $\sigma$. For an occupied group $g$, $\sum_{i \in g} |\tilde{w}_i| = \sum_{i \in g} |t_i| / \big( \sum_{j \in g} |t_j| \cdot N_{q\sigma} \big) = 1/N_{q\sigma}$, which proves (b); the per-event scale $s_i$ is inside $t_i$ and cancels in this ratio, so it moves weight only within a group. Summing over the $N_{q\sigma}$ groups on a side gives $\sum_{i \in q\sigma} |\tilde{w}_i| = 1$ for both sides. Every weight on a side has the same sign, so the long gross equals the short gross, and the global rescaling in (9) multiplies all weights by one positive constant, which proves (a). For (c), industry $k$'s long gross is the sum over its occupied long groups, $n_k^+ / N_{q+}$, and likewise on the short side. Net exposure is the difference, and it vanishes when $n_k^+ / N_{q+} = n_k^- / N_{q-}$, in particular when both sides occupy the same set of groups. The size argument is symmetric. □

**Proof of Proposition 6.** Let the notional of position $i$ be $n_i = \min(|w_i|\, bV,\, \ell_i)$, where $\ell_i$ is its liquidity limit. Position $i$ stops absorbing marginal capital at $b_i = \ell_i / (|w_i| V)$, which increases in liquidity. Write $\mathcal{U}(b) = \{i : b < b_i\}$ for the uncapped set, which shrinks as $b$ grows. The marginal alpha per marginal gross dollar is the $|w|$-weighted average of $\alpha_i$ over $\mathcal{U}(b)$. Raising $b$ removes the members with the smallest $b_i$, which are the least liquid names and, by assumption, those with the largest $\alpha_i$. The weighted average therefore cannot increase, so expected arithmetic P&L, the integral of this non-increasing marginal alpha, is concave in $b$. Volatility is non-decreasing in gross exposure, so $\mu(b) - \frac{1}{2}\sigma^2(b)$ is concave with $\mu(0) = \sigma(0) = 0$ and positive slope at zero when the uncapped book has positive alpha. As $b \to \infty$ the alpha saturates while the variance term does not, so the function eventually decreases and the maximum is interior. □

**Proof of Proposition 2.** $R_{k,s} \geq -1$ and $f \leq 0.95$ give $1 + fR_{k,s} \geq 1 - 0.95 > 0$. The second derivative is $G_k''(f) = -\sum_s p_s R_{k,s}^2/(1+fR_{k,s})^2 < 0$ unless every $R_{k,s} = 0$. Direct evaluation gives $G_k(0) = 0$ and $G_k'(0) = \sum_s p_s R_{k,s}$. Since $G_k$ is concave, its maximizer on $[0,1)$ is positive if and only if $G_k'(0) > 0$. □

**Proof of Proposition 3.** Substitute $\log(1+y) = y - \frac{1}{2}y^2 + O(y^3)$ and take expectations, which gives $G_k(f) = f\mu_k - \frac{1}{2}f^2 m_k + O(f^3 \mathbb{E}|R_k|^3)$. Maximizing the quadratic gives $f^* = \mu_k/m_k$ and $g^* = \mu_k^2/(2m_k)$. Dividing numerator and denominator by $\sigma_k^2$ gives $\frac{1}{2}\mathrm{SR}_k^2/(1+\mathrm{SR}_k^2)$, which is increasing in $\mathrm{SR}_k$ for $\mathrm{SR}_k > 0$. □

**Proof of Proposition 4.** $G_k$ is concave with $G_k(0) = 0$. If $G_k(f_{\min}) \leq 0$, concavity gives $G_k(f) \leq (f/f_{\min})\, G_k(f_{\min}) \leq 0$ for every $f \geq f_{\min}$, so no stake on the grid has positive growth and the contract cannot be selected. Selectability therefore requires $G_k(f_{\min}) > 0$. In the quadratic approximation this reads $f_{\min}\mu_k - \frac{1}{2}f_{\min}^2 m_k > 0$, which is (14). □

**Proof of Proposition 5.** (a) is immediate from (15). (b) Under independence and the quadratic approximation, portfolio growth is $\sum_i (\pi_i \mu_i - \frac{1}{2}\pi_i^2 m_i)$. Setting $\pi_i = \lambda \mu_i/m_i$ gives $\sum_i \frac{\mu_i^2}{m_i}(\lambda - \frac{1}{2}\lambda^2) = (2\lambda - \lambda^2)\sum_i g_i^*$. The portfolio mean is $\lambda \sum_i \mu_i^2/m_i$ and its standard deviation is $\lambda(\sum_i \mu_i^2\sigma_i^2/m_i^2)^{1/2}$, so $\lambda$ cancels in the ratio. With $N$ bets of comparable size, the numerator is $O(N)$ and the denominator $O(\sqrt{N})$. (c) A long option position costs its premium and has a non-negative payoff, so its loss is at most the premium, and premium outstanding is at most $cV_q$ by construction of $B_q$. □

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Table 5: The equity family in lineage order. "Unit / cap" is the base-unit fraction and the gross-exposure cap (per entry cohort). Returns, Sharpe ratios and drawdowns are from `results/<name>/summary.json`.

| Book | Status | What it does | Unit / cap | Return | Sharpe | Max DD |
|---|---|---|---|---|---|---|
| `extreme_picks` | validated | Buy the top 10% of earnings surprises, short the bottom 10%, equal bets | 0.003 / 1.5× | 4.9% | 0.93 | −18.5% |
| `graded_bets` | validated | Bet on every stock; the more extreme the surprise, the bigger the bet | 0.002 / 1.5× | 3.9% | 1.01 | −14.0% |
| `size_balanced` | validated | Extreme Picks, with an equal budget for each of the five company sizes | 0.003 / 1.5× | 4.7% | 0.91 | −17.5% |
| `conviction_tilt` | validated | Skip the middle 15% either side of average, lean toward the strongest surprises, balance by size | 0.002 / 1.5× | 5.2% | 1.18 | −13.1% |
| `sector_balanced` | validated | Conviction Tilt, also balanced across 12 industry groups; conservative 1.5x borrowing limit | 0.0015 / 1.5× | 4.1% | 1.33 | −10.8% |
| `balanced_leverage` | validated | Sector Balanced, allowed to hold up to 2.5x the account in positions | 0.003 / 2.5× | 6.5% | 1.30 | −16.6% |
| `market_boost` | validated | Balanced Leverage plus a market position worth half the account | 0.003 / 2.5× | 10.1% | 0.97 | −23.2% |
| `market_exposed` | reference | Balanced Leverage signal, longs and shorts not forced to match, market moves left in | 0.003 / 2.5× | 7.3% | 1.11 | −15.5% |
| `calm_weighted` | validated | Balanced Leverage, with each bet divided by how jumpy the stock is | 0.003 / 2.5× | 6.7% | 1.47 | −15.6% |
| `calm_weighted_matched` | validated | Calm Weighted resized so its ups and downs match Balanced Leverage exactly (4.94% a year) | 0.00346881 / 2.5× | 7.3% | 1.45 | −17.0% |
| `volatility_filter` | research | Balanced Leverage, skipping long/short x jumpiness groups that have not made money so far | 0.003 / 2.5× | 6.3% | 1.23 | −16.7% |
| `calm_weighted_filter` | research | Calm Weighted plus the Volatility Filter | 0.003 / 2.5× | 6.5% | 1.41 | −15.7% |
| `calm_rank_tilt` | research | Balanced Leverage, with bets shrunk sharply for the jumpiest stocks (ranked against last quarter) | 0.003 / 2.5× | 7.1% | 1.81 | −11.6% |
| `calm_rank_levered` | research | Calm Rank Tilt with its lower risk spent on larger bets (about 1.7x) and a 4x borrowing limit | 0.005 / 4× | 9.2% | 1.73 | −15.9% |

Table 6: Equity headline books, 1996–2013 (net of modeled costs; borrow and financing not modeled).

| Book | Return | Vol | Sharpe | Max DD | Calmar | Role |
|---|---|---|---|---|---|---|
| calm_weighted | 6.71% | 4.49% | 1.47 | −15.6% | 0.43 | validated headline |
| calm_rank_tilt | 7.12% | 3.83% | 1.81 | −11.6% | 0.61 | research headline |
| balanced_leverage | 6.49% | 4.94% | 1.30 | −16.6% | 0.39 | reference |
| market_exposed | 7.30% | 6.56% | 1.11 | −15.5% | 0.47 | reference |
| market_boost | 10.08% | 10.51% | 0.97 | −23.2% | 0.43 | validated |

Table 7: Leveraged and market-exposed research variants (top block) with the unlevered books for comparison (bottom block). Average and maximum gross exposure divided by NAV are measured on days with positions, from daily.csv.gz; ADV-capped is the share of positions at the 5% volume cap.

| Book | Return | Vol | Sharpe | Max DD | Avg gross | Max gross | ADV-capped | Beta |
|---|---|---|---|---|---|---|---|---|
| market_boost_max | 15.0% | 13.7% | 1.09 | −37.0% | 3.6× | 6.9× | 41% | 0.48 |
| balanced_leverage_max | 12.1% | 12.2% | 1.00 | −36.5% | 4.5× | 7.6× | 41% | −0.01 |
| sector_balanced_max | 12.1% | 12.2% | 1.00 | −36.5% | 4.5× | 7.6× | 41% | −0.01 |
| market_exposed_max | 11.8% | 11.8% | 1.00 | −33.0% | 2.8× | 5.0× | 26% | 0.41 |
| graded_bets_max | 10.8% | 12.4% | 0.89 | −38.8% | 4.1× | 7.2× | 26% | 0.02 |
| conviction_tilt_max | 10.7% | 12.3% | 0.89 | −40.0% | 3.5× | 5.9× | 32% | 0.02 |
| extreme_picks_max | 9.5% | 12.2% | 0.80 | −36.2% | 2.5× | 4.5× | 52% | 0.05 |
| size_balanced_max | 8.4% | 12.3% | 0.72 | −37.5% | 2.5× | 4.4× | 47% | 0.00 |
| balanced_leverage | 6.5% | 4.9% | 1.30 | −16.6% | 1.8× | 2.9× | 11% | 0.00 |
| calm_weighted | 6.7% | 4.5% | 1.47 | −15.6% | 1.8× | 2.8× | 11% | 0.00 |
| calm_rank_tilt | 7.1% | 3.8% | 1.81 | −11.6% | 1.6× | 2.7× | 11% | −0.00 |
| calm_rank_levered | 9.2% | 5.2% | 1.73 | −15.9% | 2.3× | 3.8× | 19% | −0.01 |
| market_boost | 10.1% | 10.5% | 0.97 | −23.2% | 1.3× | 2.2× | 11% | 0.51 |

Table 8: Annualized alpha against the CRSP value-weighted market, with and without a size factor, by era. $t$-statistics use Newey–West errors; IR is alpha over residual volatility. Source: results/alpha_benchmark.csv.

| Book | Return | Alpha | $t$ | Beta | Alpha (+size) | $t$ | IR | 96–01 | 02–07 | 08–13 |
|---|---|---|---|---|---|---|---|---|---|---|
| calm_rank_tilt | 7.1% | 7.3% | 7.6 | −0.00 | 7.2% | 7.4 | 1.86 | 12.4% | 7.1% | 3.1% |
| calm_weighted | 6.7% | 6.9% | 6.1 | 0.00 | 6.9% | 6.1 | 1.51 | 11.1% | 7.0% | 3.0% |
| balanced_leverage | 6.5% | 6.7% | 5.3 | 0.00 | 6.7% | 5.4 | 1.32 | 10.5% | 7.2% | 2.7% |
| conviction_tilt | 5.2% | 5.4% | 4.7 | 0.01 | 5.2% | 4.6 | 1.19 | 8.2% | 4.7% | 3.1% |
| sector_balanced | 4.1% | 4.2% | 5.5 | 0.00 | 3.9% | 5.3 | 1.35 | 6.6% | 4.0% | 2.2% |
| extreme_picks | 4.9% | 5.2% | 4.0 | −0.00 | 6.5% | 5.3 | 0.95 | 6.2% | 6.6% | 2.3% |
| size_balanced | 4.7% | 5.0% | 3.9 | 0.00 | 6.1% | 4.9 | 0.93 | 6.8% | 5.3% | 2.5% |
| graded_bets | 3.9% | 4.1% | 4.1 | 0.00 | 4.8% | 4.9 | 1.03 | 5.3% | 4.8% | 1.8% |
| market_exposed | 7.3% | 5.8% | 4.0 | 0.18 | 6.7% | 4.7 | 1.02 | 10.1% | 6.6% | 1.0% |
| market_boost | 10.1% | 4.4% | 4.9 | 0.51 | 4.4% | 5.0 | 1.21 | 7.4% | 5.0% | 1.7% |
| calm_rank_levered | 9.2% | 9.5% | 7.3 | −0.01 | 9.5% | 7.2 | 1.80 | 16.8% | 9.8% | 2.8% |
| balanced_leverage_max | 12.1% | 13.0% | 4.2 | −0.01 | 14.3% | 4.6 | 1.04 | 19.9% | 18.3% | 0.4% |
| market_boost_max | 15.0% | 9.5% | 3.8 | 0.48 | 10.7% | 4.4 | 0.94 | 14.8% | 15.2% | −0.5% |

Table 9: Alpha per dollar traded on the long and short legs under the two benchmarks, and the size-matched dollar alpha of each leg. Source: `results/situations_all.csv`, `situations_books.csv`, `profiles_timing.csv`.

| | All-announcer benchmark (bp per $) | | | Size-matched benchmark (bp per $) | | | Size-matched alpha | |
|---|---|---|---|---|---|---|---|---|
| Book | Long | Short | L/S | Long | Short | L/S | Long $ | Short $ |
| extreme_picks | 103 | 52 | 2.0 | 140 | 54 | 2.6 | $12.5M | $4.5M |
| graded_bets | 64 | 57 | 1.1 | 92 | 59 | 1.6 | $8.0M | $5.0M |
| size_balanced | 131 | 65 | 2.0 | 165 | 52 | 3.2 | $12.6M | $4.2M |
| conviction_tilt | 112 | 62 | 1.8 | 124 | 62 | 2.0 | $12.4M | $6.5M |
| sector_balanced | 125 | 57 | 2.2 | 141 | 55 | 2.6 | $9.2M | $3.7M |
| balanced_leverage | 74 | 71 | 1.0 | 108 | 50 | 2.2 | $18.2M | $8.6M |
| calm_weighted | 74 | 78 | 0.9 | 108 | 55 | 2.0 | $18.4M | $9.6M |
| calm_rank_tilt | 65 | 96 | 0.7 | 105 | 65 | 1.6 | $18.2M | $11.6M |
| market_exposed | 57 | 68 | 0.9 | 96 | 54 | 1.8 | $17.8M | $10.0M |

Table 10: The recipe "long, thin, low-beta, held over 60 days" in each book (size-matched alpha). "Eras +" is the number of the three eras in which its alpha is positive; "Top recipe" says whether it is the book's most profitable recipe. Source: `results/profiles_recipes.csv`.

| Book | Money | Alpha share | bp | $t$ | Quarters + | Eras + | Real $ | Top recipe |
|---|---|---|---|---|---|---|---|---|
| extreme_picks | 2.2% | 10.8% | 490 | 4.8 | 71% | 3/3 | $2.8M | yes |
| graded_bets | 2.2% | 15.0% | 522 | 6.5 | 76% | 3/3 | $3.2M | yes |
| size_balanced | 2.4% | 11.9% | 529 | 4.9 | 68% | 3/3 | $3.0M | no |
| conviction_tilt | 2.9% | 17.0% | 539 | 5.9 | 76% | 3/3 | $5.1M | yes |
| sector_balanced | 3.1% | 20.7% | 648 | 6.6 | 78% | 3/3 | $3.9M | yes |
| balanced_leverage | 2.1% | 14.1% | 530 | 5.8 | 76% | 3/3 | $5.8M | yes |
| calm_weighted | 2.1% | 13.0% | 489 | 5.4 | 75% | 2/3 | $5.7M | yes |
| calm_rank_tilt | 2.1% | 10.1% | 406 | 4.1 | 71% | 2/3 | $5.0M | yes |
| market_exposed | 1.9% | 13.1% | 527 | 5.7 | 76% | 2/3 | $8.0M | yes |

Table 11: High-beta shorts, low/mid-beta shorts and high-beta longs in each book: alpha per dollar (bp), quarter-clustered $t$, real profit and loss, and for high-beta shorts the number of eras with a real-dollar loss. Source: `results/profiles_beta_legs.csv`.

| | High-beta shorts | | | | Low/mid-beta shorts | | | High-beta longs | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Book | bp | $t$ | Real $ | Losing eras | bp | $t$ | Real $ | bp | $t$ | Real $ |
| extreme_picks | −40 | −0.5 | −$6.1M | 3/3 | 108 | 3.4 | −$1.4M | 35 | 0.9 | $4.3M |
| graded_bets | 27 | 0.3 | −$4.3M | 3/3 | 76 | 2.7 | −$4.7M | 23 | 0.8 | $4.7M |
| size_balanced | −41 | −0.6 | −$5.6M | 3/3 | 101 | 3.4 | −$1.8M | 23 | 1.0 | $3.7M |
| conviction_tilt | 29 | 0.3 | −$5.2M | 3/3 | 80 | 2.8 | −$5.2M | 28 | 0.9 | $6.0M |
| sector_balanced | 1 | −0.0 | −$3.6M | 3/3 | 81 | 2.6 | −$3.3M | 73 | 0.7 | $4.3M |
| balanced_leverage | 6 | 0.1 | −$8.3M | 3/3 | 73 | 2.6 | −$7.1M | 52 | 0.5 | $9.2M |
| calm_weighted | 17 | 0.1 | −$6.8M | 3/3 | 71 | 2.4 | −$7.6M | 61 | 0.8 | $8.7M |
| calm_rank_tilt | 47 | 0.0 | −$3.3M | 2/3 | 71 | 2.2 | −$7.5M | 71 | 1.3 | $6.6M |
| market_exposed | 16 | 0.4 | −$21.6M | 3/3 | 74 | 2.6 | −$29.3M | 50 | 0.7 | $23.0M |

Table 12: Selected situations for `balanced_leverage`, all-announcer benchmark. Money and alpha are shares of the book; $t$ is clustered by quarter; verdicts are defined in `docs/SITUATIONS.md`.

| Cut | Situation | Money | Alpha | bp/$ | $t$ | Verdict |
|---|---|---|---|---|---|---|
| Company size | size 1 (smallest) | 9% | 25% | 203 | 3.3 | ENGINE |
| Company size | size 2 | 19% | 26% | 102 | 3.7 | ENGINE |
| Company size | size 3 | 23% | 25% | 78 | 4.5 | ENGINE |
| Company size | size 4 | 25% | 15% | 43 | 3.0 | WORKS |
| Company size | size 5 (largest) | 25% | 10% | 28 | 2.4 | WORKS |
| How easily traded | liquidity 1 (thinnest) | 7% | 23% | 249 | 5.9 | ENGINE |
| How easily traded | liquidity 2 | 20% | 27% | 99 | 3.9 | ENGINE |
| How easily traded | liquidity 3 | 24% | 27% | 82 | 4.1 | ENGINE |
| How easily traded | liquidity 4 | 25% | 16% | 47 | 3.0 | WORKS |
| How easily traded | liquidity 5 (most traded) | 24% | 7% | 22 | 1.5 | UNPROVEN |
| Stock's market sensitivity | low–beta stocks | 31% | 55% | 129 | 6.6 | ENGINE |
| Stock's market sensitivity | mid–beta stocks | 34% | 36% | 76 | 3.5 | ENGINE |
| Stock's market sensitivity | high–beta stocks | 33% | 9% | 20 | −0.0 | UNPROVEN |
| Hold length | ¡=40d | 38% | 19% | 37 | 2.5 | WORKS |
| Hold length | 41–60d | 23% | 13% | 42 | 2.0 | WORKS |
| Hold length | 61–90d | 29% | 26% | 65 | 2.7 | WORKS |
| Hold length | ¿90d | 10% | 42% | 303 | 5.0 | ENGINE |
| Period | 1996–01 | 21% | 38% | 136 | 5.7 | ENGINE |
| Period | 2002–07 | 33% | 41% | 89 | 3.5 | ENGINE |
| Period | 2008–13 | 46% | 21% | 33 | 1.2 | UNPROVEN |

Table 13: Alpha per dollar traded (bp, size-matched benchmark; quarter-clustered $t$ in parentheses) for `balanced_leverage` positions cut by beta tercile and idiosyncratic-volatility tercile. The right column and bottom row are the margins. Positions are read from `results/balanced_leverage/positions.parquet`.

| | Calm | Mid-vol | Jumpy | All volatilities |
|---|---|---|---|---|
| Low-beta | 128 (7.6) | 128 (4.6) | 189 (4.1) | 136 (7.5) |
| Mid-beta | 61 (2.8) | 80 (3.7) | 88 (0.9) | 75 (3.6) |
| High-beta | 33 (−0.2) | 63 (2.6) | 9 (−0.3) | 29 (0.4) |
| All betas | 91 (7.0) | 88 (5.4) | 57 (2.0) | |

Table 14: Kelly options books. The top block uses the corrected convention (dead quotes marked at half the ask, or zero in the pessimistic bound, with a one-tick cost floor); the bottom block is the *overstated* legacy convention, shown only so the size of the correction is visible. Source: `results/<name>/summary.json`.

| Book | Dead-quote mark | Return | Vol | Sharpe | Max DD | Calmar | Role |
|---|---|---|---|---|---|---|---|
| `options_cap20` | zero_bid_half_ask | 4.12% | 2.30% | 1.77 | −13.8% | 0.30 | validated headline |
| `options_cap30` | zero_bid_half_ask | 5.69% | 3.61% | 1.55 | −22.3% | 0.26 | validated |
| `options_worst_case` | zero_bid_zero | 2.34% | 2.37% | 0.99 | −17.1% | 0.14 | pessimistic bound |
| `options_cap20_legacy` | carry_last | 8.58% | 2.18% | 3.78 | −10.3% | 0.83 | overstated |
| `options_cap30_legacy` | carry_last | 12.26% | 3.33% | 3.49 | −17.4% | 0.70 | overstated |
| `options_cap40_legacy` | carry_last | 15.79% | 4.50% | 3.28 | −23.9% | 0.66 | overstated |

Table 15: The two sleeves and the three blends, with Sharpe ratios by era. Source: `results/<name>/summary.json`.

| Book | Role | Return | Vol | Sharpe | Max DD | Sh 96–01 | Sh 02–07 | Sh 08–13 |
|---|---|---|---|---|---|---|---|---|
| `calm_weighted` | equity sleeve | 6.71% | 4.49% | 1.47 | −15.6% | 2.40 | 1.63 | 0.59 |
| `options_cap20` | options sleeve | 4.12% | 2.30% | 1.77 | −13.8% | 0.69 | 2.04 | 2.39 |
| `hedged_blend` | validated | 11.33% | 4.83% | 2.25 | −16.7% | 2.41 | 2.70 | 1.91 |
| `calm_blend` | research | 12.43% | 4.73% | 2.50 | −15.5% | 2.92 | 2.84 | 2.10 |
| `legacy_blend` | overstated | 14.25% | 4.18% | 3.21 | −9.9% | 2.79 | 4.64 | 2.61 |

Table 16: Calendar-year returns (%) of `calm_weighted`, `options_cap20` and the market. The options book has no trades before 1998 and no data for 2011 (the source file is corrupt), and its 2013 figure covers January to August; dashes mark years without trades.

| Year | Equity | Options | Market | Year | Equity | Options | Market |
|---|---|---|---|---|---|---|---|
| 1996 | 14.0% | – | 21.2% | 2005 | 8.5% | −0.5% | 7.3% |
| 1997 | 14.4% | – | 30.4% | 2006 | 4.9% | 4.2% | 16.2% |
| 1998 | 14.8% | 2.9% | 22.3% | 2007 | 12.9% | 9.3% | 7.3% |
| 1999 | 12.0% | −1.2% | 25.7% | 2008 | 10.6% | −10.0% | −38.2% |
| 2000 | 11.3% | 4.0% | −11.3% | 2009 | −6.9% | 6.1% | 31.2% |
| 2001 | 6.8% | −0.3% | −11.0% | 2010 | 2.6% | 9.2% | 17.7% |
| 2002 | 5.7% | −6.1% | −20.9% | 2011 | 4.0% | – | −1.1% |
| 2003 | 2.5% | 18.8% | 33.1% | 2012 | 1.8% | −1.7% | 15.8% |
| 2004 | 7.6% | 12.3% | 13.0% | 2013 | 5.5% | 27.8% | 30.5% |

Table 17: Equity capacity frontier: sweep over the base-unit fraction $b$ at a six-times gross cap (all other parameters fixed); the row used for `balanced_leverage_max` is in bold. Source: `data/sweep_aggressive_strategy6.csv`.

| $b$ ($\times 10^{-3}$) | Return | Vol | Sharpe | Max DD | ADV-capped | Avg gross / initial cap. |
|---|---|---|---|---|---|---|
| 4.5 | 8.16% | 6.52% | 1.24 | −22.0% | 11.6% | 7.1× |
| 6 | 9.24% | 7.90% | 1.16 | −26.6% | 15.5% | 10.4× |
| 7.5 | 10.12% | 9.11% | 1.10 | −30.5% | 19.1% | 14.0× |
| 9 | 10.89% | 10.23% | 1.06 | −33.3% | 22.3% | 17.7× |
| 10.5 | 11.57% | 11.27% | 1.03 | −35.3% | 25.2% | 21.4× |
| **12** | **12.15%** | **12.20%** | **1.00** | **−36.5%** | **27.9%** | **24.7×** |
| 15 | 10.63% | 13.72% | 0.81 | −43.2% | 36.7% | 22.9× |
| 18 | 10.43% | 14.09% | 0.77 | −41.0% | 45.2% | 21.5× |
| 21 | 8.27% | 15.25% | 0.60 | −45.0% | 51.8% | 15.6× |
| 24 | 6.85% | 16.31% | 0.49 | −50.7% | 55.6% | 12.7× |
| 30 | 1.83% | 19.35% | 0.19 | −64.0% | 60.5% | 7.3× |
| 36 | 0.39% | 22.14% | 0.13 | −66.2% | 62.7% | 5.6× |

Table 18: Probabilistic and deflated Sharpe ratios on active-period daily returns. $N$ is the counted number of trials; $\widehat{\mathrm{SR}}_0$ is the expected maximum Sharpe ratio under the null for that $N$ and the dispersion of the counted trials; $N^*$ is the number of trials at which the DSR would fall below 0.95 (same dispersion).

| Book | $N$ | Active Sharpe | $\widehat{\mathrm{SR}}_0$ | PSR vs 0 | DSR | $N^*$ for DSR$< 0.95$ |
|---|---|---|---|---|---|---|
| calm_weighted | 326 | 1.51 | 0.71 | 1.000 | 1.00 | 251,189 |
| calm_rank_tilt | 326 | 1.86 | 0.71 | 1.000 | 1.00 | $> 10^6$ |
| balanced_leverage | 326 | 1.33 | 0.71 | 1.000 | 1.00 | 10,680 |
| balanced_leverage_max | 326 | 1.03 | 0.71 | 1.000 | 0.91 | 139 |
| options_cap20 | 30 | 1.89 | 2.70 | 1.000 | 0.00 | 5 |
| options_cap30 | 30 | 1.65 | 2.70 | 1.000 | 0.00 | 4 |
| hedged_blend | 326 | 2.31 | 0.71 | 1.000 | 1.00 | $> 10^6$ |
| calm_blend | 326 | 2.57 | 0.71 | 1.000 | 1.00 | $> 10^6$ |

Table 19: Deflated Sharpe ratio as the assumed number of independent trials $N$ grows (dispersion of the counted trials held fixed).

| Book | $N = 10$ | $N = 30$ | $N = 100$ | $N = 300$ | $N = 1{,}000$ | $N = 10{,}000$ |
|---|---|---|---|---|---|---|
| calm_weighted | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.99 |
| calm_rank_tilt | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 |
| options_cap20 | 0.26 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| options_cap30 | 0.06 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |

Table 20: Defects found. "Before → after" is return / Sharpe ratio where `results/` contains both runs; rows marked *identified* are not yet applied to the numbers reported here.

| Defect | What exposed it | Effect (before → after) |
|---|---|---|
| Options quotes carried forward | Dead options (zero bid) were marked at their last live mid, so a dying contract was never marked down | 20% book: 8.58% / 3.78 → 4.12% / 1.77; 30% book: 12.26% / 3.49 → 5.69% / 1.55 |
| Benchmark artifact in the long/short split | The all-announcer benchmark is biased when side and size are mixed; it made the short leg look as good as the long leg | Long-to-short ratio 0.7 to 2.2 (all-announcer) → 1.6 to 3.2 (size-matched) |
| Leverage cap not binding on carried-over positions | The cap applies to each entry cohort, so open-book gross exceeds the nominal cap | `calm_weighted`: cap 2.5 ×, peak 2.8 ×, over the cap on 4.0% of days. identified; not yet applied to the reported numbers |
| Decile look-ahead | Deciles are ranked on the full announcement quarter's cross-section, so an early announcer is ranked against later ones | *Identified*: identified; not yet applied to the reported numbers |
| Full-sample hold-period fitting | Decay-day windows are estimated on the same history they are tested on | *Identified*: identified; not yet applied to the reported numbers |
| Sharpe denominator | Daily P&L was once divided by initial capital instead of prior-day NAV, overstating volatility as the NAV compounded | `calm_weighted`: 1.15 (fixed capital) → 1.47; `options_cap20`: 1.91 → 1.77 |
| EAR leakage | The day-0-to-day-1 return overlaps the first day of held P&L; the Sharpe ratio rose with the weight on it | Sharpe ratio flat once removed; EAR not adopted (project history) |
| Ambient beta in the return law | The selector chose calls even in D1, since 1996–2013 was a net up market | Return law built from market-adjusted returns, recentered on a trailing drift (project history) |
| Strike units | OptionMetrics strikes were stored ×1000, so puts dominated even D10 at near-full Kelly fractions | Strikes divided by 1000 at source (project history) |
| Full-sample return law | Early trades were priced with later data | Walk-forward law after an eight-quarter warm-up (project history) |
| Independent Kelly sizing | The premium budget was exhausted by a few bets, which funded only a handful of positions | Kelly-proportional allocation of a shared budget (project history) |
| Per-cohort premium cap | Open premium could exceed the cap across overlapping 60-day cohorts | Cap on total outstanding premium (project history) |
| Data drift on a staged copy | Decay-day cells changed with no code change when a staged data copy was used | Results restored to the canonical data; checksum manifest and a pre-commit hook; the build pins pandas 3 |

Table 21: Mapping from the numbered labels used in earlier drafts and the project history to the plain names used in this paper. The rename retired the numbered labels and left every number unchanged.

| Label in earlier drafts | Plain name here |
|---|---|
| Strategy 1 (extreme-decile L/S) | `extreme_picks` |
| Strategy 2 (rank-weighted) | `graded_bets` |
| Strategy 3 (balanced) | `size_balanced` |
| Strategy 4 | `conviction_tilt` |
| Strategy 5 | `sector_balanced` |
| Strategy 6 | `balanced_leverage` |
| Strategy 6 + 0.5x beta overlay | `market_boost` |
| Strategy 7 | `market_exposed` |
| Strategy 9a | `volatility_filter` |
| Strategy 9b | `calm_weighted` |
| Strategy 9c | `calm_weighted_filter` |
| Strategy 10 | `calm_rank_tilt` |
| Strategy 6 / 5 / 4 ... Aggressive | balanced_leverage_max, sector_balanced_max, ... (the *_max books) |
| Strategy I (earlier drafts' equity headline) | balanced_leverage_max (= sector_balanced_max) |
| Strategy II, opt_kelly_cap30 (earlier drafts' options headline) | options_cap30_legacy (overstated); corrected: options_cap30 |
| opt_kelly_cap20 | options_cap20_legacy (overstated); corrected: options_cap20 |
| opt_kelly_cap40 | `options_cap40_legacy` |
| Strategy 8 (walk-forward ML) | not in the framework; excluded (ran on an unverified data copy) |

Table 22: Equity parameters (from each run's `summary.json`). Base unit is the fraction of trailing NAV per unit of weight; the gross cap applies per entry cohort.

| Book | Base unit | Gross cap | ADV cap | Returns | Cap mode | Overlay |
|---|---|---|---|---|---|---|
| `extreme_picks` | 0.003 | 1.5× | 5% | mktadj | proportional | 0 |
| `graded_bets` | 0.002 | 1.5× | 5% | mktadj | proportional | 0 |
| `size_balanced` | 0.003 | 1.5× | 5% | mktadj | proportional | 0 |
| `conviction_tilt` | 0.002 | 1.5× | 5% | mktadj | priority | 0 |
| `sector_balanced` | 0.0015 | 1.5× | 5% | mktadj | priority | 0 |
| `balanced_leverage` | 0.003 | 2.5× | 5% | mktadj | priority | 0 |
| `market_boost` | 0.003 | 2.5× | 5% | mktadj | priority | 0.5 |
| `market_exposed` | 0.003 | 2.5× | 5% | raw | priority | 0 |
| `calm_weighted` | 0.003 | 2.5× | 5% | mktadj | priority | 0 |
| `calm_rank_tilt` | 0.003 | 2.5× | 5% | mktadj | priority | 0 |
| `calm_rank_levered` | 0.005 | 4× | 5% | mktadj | priority | 0 |
| `balanced_leverage_max` | 0.012 | 6× | 5% | mktadj | priority | 0 |
| `market_boost_max` | 0.012 | 6× | 5% | mktadj | priority | 0.5 |

Table 23: Options parameters (from each run's `summary.json`).

| Book | Premium cap | Kelly mult. | Hold (d) | Round-trip | Floor / share | Dead-quote mark |
|---|---|---|---|---|---|---|
| options_cap20 | 0.2 | 0.5 | 60 | 3% | $0.05 | zero_bid_half_ask |
| options_cap30 | 0.3 | 0.5 | 60 | 3% | $0.05 | zero_bid_half_ask |
| options_worst_case | 0.2 | 0.5 | 60 | 3% | $0.05 | zero_bid_zero |
| options_cap20_legacy | 0.2 | 0.5 | 60 | 3% | $0.00 | carry_last |
| options_cap30_legacy | 0.3 | 0.5 | 60 | 3% | $0.00 | carry_last |
| options_cap40_legacy | 0.4 | 0.5 | 60 | 3% | $0.00 | carry_last |